

Command words

What each one asks for, how the marks are given, and what the examiners say goes wrong

Write in full sentences, make every point complete, and answer the question that was asked about the scenario that was given.

WHICH WORDS CARRY THE MARKS

Every Paper 1 and Paper 3 question part from 2022 to March 2026: 925 parts in 56 papers. Each part counts under the command word it uses.

Word	Cambridge's definition (syllabus p. 66)	Parts	Share of marks	
Describe	state the points of a topic/give characteristics and main features	451	41.0%	
Explain	set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence	197	18.6%	
Discuss	write about issue(s) or topic(s) in depth in a structured way	80	15.4%	
Evaluate	judge or calculate the quality, importance, amount, or value of something	25	4.8%	
Analyse	examine in detail to show meaning, identify elements and the relationship between them	19	3.4%	
Justify	support a case with evidence/argument	20	3.1%	
Complete	not in the syllabus list; an instruction, not a command word	17	3.0%	
Identify	name/select/recognise	28	2.6%	
Compare	identify/comment on similarities and/or differences	16	2.3%	
Draw	not in the syllabus list; an instruction, not a command word	9	1.2%	
Write	not in the syllabus list; an instruction, not a command word	9	1.0%	
Give	not in the syllabus list; an instruction, not a command word	15	1.0%	
State	express in clear terms	13	0.8%	
Contrast	identify/comment on differences	9	0.7%	

“WRITE EVERYTHING IN BULLET POINTS” IS WRONG

1. No Cambridge document for 9626 tells you to answer in bullet points. Cambridge's own guide for students says the opposite.

“Answer the questions in sentences. Never give ‘one word’ or one ‘phrase’ answers.” Cambridge Learner Guide for 9626, p. 15

2. The examiners repeat this in almost every series. Describe and Explain answers need full sentences too, not only the long questions.

"Candidates should also be encouraged to write full sentences and discouraged from writing bulleted, short statements in their responses. Descriptions or explanations can only be conveyed in full sentences." Examiner report, November 2025, Paper 9626/32, p. 16

"Candidates should write full sentences and avoid bulleted short statements in their responses." Examiner report, June 2022, Paper 9626/31, p. 13

3. The real problem with bullets is what they leave out. A bullet is usually a bare point with no reason or detail, and the reason or detail is where most marks are.

"the use of bullet points rarely produces little more than simple points or short statements with no explanations or descriptions" Examiner report, June 2018, Paper 9626/32, p. 17

"Short statements or bulleted responses that do not expand on facts cannot be awarded the higher marks because these are not discussions." Examiner report, November 2022, Paper 9626/31, p. 14

4. For Discuss, Evaluate and Analyse, the shape of the answer matters as well. Bulleted lists and tables of statements lose most of the marks, and so do separate lists of benefits and drawbacks when each item is only a word or a very brief sentence.

"learners who produced bulleted lists or tables of statements were unable to access most of the marks" Examiner report, March 2025, Paper 9626/32, p. 10

"if these answers are shown in a sketched table, they do not properly answer the question as there is no e.g., discussion" Examiner report, March 2025, Paper 9626/32, p. 8

"This comment also applies to answers that have separate lists of benefits and drawbacks, but the lists consist of single word or very brief sentences." Examiner report, June 2025, Paper 9626/32, p. 22

5. This was once written into the mark schemes too. From 2021 to 2023, almost every Paper 1 and Paper 3 mark scheme printed a cap on bulleted answers to its Discuss, Evaluate or Analyse questions: at most 6 of 8 marks (fewer on shorter questions, e.g. 3 of 4), however good the points were. In 2024 only the March and June Paper 3 mark schemes still printed it, and from November 2024 on it is gone. The 2025 examiner reports still say bulleted lists and tables cannot reach most of the marks.

"Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion" Mark scheme 9626/12, March 2022, p. 4

"Max 6 marks if bullets/list of points." Mark scheme 9626/31, June 2024, p. 8

"five marks if bullets/list of points Must have expansions to be a proper analysis" Mark scheme 9626/12, March 2021, p. 9

6. Bullets are not banned outright. On one Paper 1 question (November 2025) that asked for the steps of planning something, the examiners accepted either a list of steps (sometimes bulleted) or a description, as long as each step showed a decision being made. Treat that as a rare exception, not a licence: the reports ask for full sentences everywhere else.

"In the first, candidates listed (sometimes by creating a bulleted list) steps that would be taken, whilst others described the steps that had been taken. Either method was acceptable" Examiner report, November 2025, Paper 9626/12, p. 7

How the marks are given

1. You do not lose marks for grammar or spelling. Examiners give marks for correct points and never take marks away for mistakes. Your meaning still has to be clear.

"marks are not deducted for errors" Mark scheme 9626/12, June 2025, p. 2

"answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous." Mark scheme 9626/12, June 2025, p. 2

2. Short sentences are fine. One clear point per sentence is easier for the examiner to mark and easier for you to check.

"Keep most sentences short and focused. That will help with clarity and in making every point stand out for the examiner." Cambridge Learner Guide for 9626, p. 15

3. Some mark schemes give one mark for a point and a second mark only when you add its reason or detail. One mark scheme says outright that an answer made only of bare points is capped.

"MAX two marks if answers are all only first marks." Mark scheme 9626/13, November 2023, p. 4

4. Answer the question as it is written, about the scenario it gives. Writing everything you know about a word you spotted in the question does not earn the marks.

"the full range of marks is only available to candidates for answers referring to the scenario in the questions" Examiner report, November 2025, Paper 9626/32, p. 16

"Centres must ensure that candidates target their answers to the questions as they are written and not at 'key' words in the questions." Examiner report, June 2024, Paper 9626/31, p. 17

5. Do not copy mark scheme bullets you have memorised. Examiners notice, and a list of learnt points is not an answer to a new question.

"Examiners saw many scripts where candidates listed sets of bullet points that appeared to have been learnt directly from previous mark schemes." Examiner report, November 2019, Paper 9626/11, p. 1

6. The number of marks is a clue to how much to write: how many separate points, or how much detail, the examiner expects.

"Look at how many marks are allocated to a question. This gives you a clue as to how much work is expected." Cambridge Learner Guide for 9626, p. 15

"the number of marks available for each question is a useful indicator of how much detail or how many descriptions should be given in the answers" Examiner report, March 2025, Paper 9626/32, p. 8

7. Never leave a question blank. A few sentences on a topic you are unsure of can still earn marks.

"Writing a few sentences, even if little is known about the topic, may gain valuable marks towards the overall total." Examiner report, November 2025, Paper 9626/32, p. 16

Describe

Describe — Cambridge's definition: "state the points of a topic/give characteristics and main features" Syllabus 2025–2027, p. 66

Say what it is, what it does or how it happens, in enough separate, accurate points to match the marks; you only need to say why if the question asks for reasons.

HOW THE MARKS ARE GIVEN

1. In most Describe mark schemes each correct point earns one mark. So a 4-mark Describe question needs about four separate points, not one long sentence that says one thing.

"As a description is generally a one-mark answer, candidates had to give three separate descriptions for full marks to be awarded." Examiner report, November 2025, Paper 9626/11, p. 1

"Typically, one would expect a four mark question to require four individual points to be made." Examiner report, November 2022, Paper 9626/12, p. 4

"Question 1 has four marks available for describing the structure of a cellular network so at least four points need to be made in the answer for full marks" Examiner report, November 2025, Paper 9626/33, p. 19

2. Some mark schemes give a mark for a point, shown as (1st), and a second mark for adding detail to that same point, shown as (1). On those questions you can often reach 4 marks with four short points or with two points that you expand. 83 of the 387 Describe mark schemes from 2022 on use (1st) somewhere; none before 2022 do.

"A wind sensor/anemometer (1st) – is used to measure the direction of the wind (1)" Mark scheme 9626/11, June 2025, p. 6

"It is interrogated by the inference engine (1st) – Using relevant rules/rules that apply (1)" Mark scheme 9626/11, November 2025, p. 5

"Therefore, they could describe four individual functions or two in some depth, or three with a mixture of techniques." Examiner report, June 2025, Paper 9626/13, p. 11

"This meant that candidates could gain full credit by either giving four descriptions or expanding on fewer descriptions." Examiner report, June 2025, Paper 9626/33, p. 26

3. Naming things is not describing them. Examiners report that answers which only name components miss the higher marks, and at least one mark scheme caps name-only answers (max 3 of 5 for naming events without descriptions). Say what each one is or does.

"Max three for naming valid events without descriptions" Mark scheme 9626/32, June 2024, p. 6

"Many responses indicated that candidates knew the names of the components but were unable to describe them in the detail required to gain the higher marks." Examiner report, June 2022, Paper 9626/13, p. 9

"candidates were unable to describe the component, without relying on simply restating the component's name." Examiner report, March 2024, Paper 9626/12, p. 2

4. Describe is about what and how. Explain adds why. You do not need a reason in a Describe answer unless the question asks for one, for example 'Describe, with reasons'. When it does, full marks are held back until you give the reasons.

"Explain set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence" Syllabus 2025–2027, p. 66

"Candidates were asked to explain the differences rather than describe, so they were expected to give reasons for why the differences exist." Examiner report, November 2023, Paper 9626/11, p. 2

"Describe, with reasons, the steps the manager takes when setting up this process." Question paper 9626/11, June 2023, p. 6

"Must have at least TWO reasons to obtain full marks" Mark scheme 9626/11, June 2023, p. 5

"For each part, candidates had to describe the process and give a reason for its use for full marks to be awarded." Examiner report, November 2025, Paper 9626/11, p. 3

HOW TO WRITE IT

1. Write descriptions as full sentences, one point per sentence. The Paper 3 reports say descriptions cannot be given in short bullets. On Paper 1, a list of steps for a 'describe the process' question was accepted, but full sentences are the safe choice on both papers.

"Descriptions or explanations can only be conveyed in full sentences." Examiner report, June 2022, Paper 9626/31, p. 13

"Candidates should also be encouraged to write full sentences and discouraged from writing bulleted, short statements in their responses." Examiner report, November 2025, Paper 9626/32, p. 16

"In the first, candidates listed (sometimes by creating a bulleted list) steps that would be taken, whilst others described the steps that had been taken. Either method was acceptable" Examiner report, November 2025, Paper 9626/12, p. 7

WHAT GOES WRONG

1. Examiners keep reporting that students mix up Describe and Explain. The opposite mistake also loses marks: if the question asks why, a description of the thing itself scores little or nothing.

"Many candidates failed to differentiate between, for example, questions where they were asked to describe and questions where they were asked to explain." Examiner report, June 2024, Paper 9626/11, p. 1

"where a question asks why a protocol or process would be used, there is usually little or no credit given for describing the protocol or process itself since this does not answer the question 'why?'." Examiner report, June 2022, Paper 9626/31, p. 13

2. Check what the question wants you to describe. 'Describe how X is set up' or 'the process of' wants the steps in order. 'Describe the features' or 'the functions' wants what X is or does. 'Describe the benefits' wants benefits, not uses. Writing about the wrong one earns nothing, even if it is true.

"Good answers detailed the steps that would be taken to set up an electronic wallet. No details of the use of the wallet were required to gain marks." Examiner report, June 2022, Paper 9626/32, p. 16

"some only identified methods of collecting the data without saying how they would be used for example, what data would be collected or who it would be collected from." Examiner report, June 2022, Paper 9626/13, p. 7

"Common mistakes included descriptions of uses, inaccurate references to network performance and to illegal activities." Examiner report, March 2025, Paper 9626/32, p. 9

"some focused on the use of the internet and remote learning, neither of which are features of a MOOC." Examiner report, November 2022, Paper 9626/31, p. 14

3. Do not repeat the question back. 'White box testing is used to test the code' or 'a sound sensor measures sound' earns no mark. Add a detail the question did not already give you.

"Further, many candidates repeated the question, writing that 'white box testing is used to test the code'. In general, statements that repeat the question are not given credit." Examiner report, June 2024, Paper 9626/32, p. 22

"Candidates should avoid simply repeating the question in their response without further expansion; for example, a humidity sensor measures humidity and a sound sensor measures sound." Examiner report, June 2022, Paper 9626/12, p. 5

4. Short vague words like 'cheaper', 'more expensive' or 'work out the costs' do not earn marks on their own. Say exactly what is cheaper or which cost is saved, and use the proper technical term, not slang.

"Some weaker responses contained responses such as 'more expensive' which is not enough at this level and statements such as this should always be expanded." Examiner report, June 2022, Paper 9626/13, p. 8

"However, where candidates stated that a specific, named, cost could be avoided, this was awarded." Examiner report, June 2025, Paper 9626/12, p. 6

"A common mistake was to give vague answers such as 'work out the costs'." Examiner report, June 2024, Paper 9626/32, p. 22

"candidates did not always focus as keenly as was required and gave vague answers that did not warrant any marks." Examiner report, November 2025, Paper 9626/11, p. 2

"The use of what may be considered slang often renders answers meaningless." Examiner report, March 2025, Paper 9626/12, p. 1

TIPS

1. Use the scenario. If the question is about a nuclear power plant, a bank's new customers or a given table of data, your points should fit that situation. A point that is true in general but wrong for the scenario loses the mark, and a prepared answer that ignores the question is kept to low marks.

"The full range of marks is only available to candidates for answers referring to the scenario in the questions." Examiner report, June 2022, Paper 9626/31, p. 13

"context also provides a restriction, as certain answers may be acceptable for one scenario, but not for another." Examiner report, November 2025, Paper 9626/11, p. 1

"Many responses suggested a form of lookup validation. This would not be acceptable, as the account numbers are for new customers and so would not exist." Examiner report, November 2023, Paper 9626/13, p. 9

"those who provided an answer they had prepared for, rather than for the question on the paper, were limited to low band marks." Examiner report, June 2022, Paper 9626/31, p. 15

2. If the question asks for a set number of things (for example 'Describe two reasons'), only the first that many answers are marked, so extra answers do not help. Joining two ideas with 'and' can make the second one count as a separate answer, so give each point its own sentence.

"Typically, where candidates join two concepts together using 'and', the concept after the word 'and' is treated as a second answer, unless the piece after the 'and' is clearly an explanation or clarification of the piece before the 'and'." Examiner report, November 2025, Paper 9626/11, p. 1

"The question clearly stated that two reasons should be described so it was not possible to score full marks without giving two descriptions." Examiner report, June 2025, Paper 9626/33, p. 26

"If, for example, the question asks for two reasons why an action may be taken, only the first two answers given by the candidate may be awarded marks." Examiner report, November 2025, Paper 9626/11, p. 1

WORKED EXAMPLE

Q4(b) [4 marks] "(b) Describe how a knowledge base is used in an expert system." Question paper 9626/11, November 2025, p. 4

Part (a) asked what is meant by the term knowledge base.

A full-mark answer, built from the mark scheme (not a real candidate's):

1. The knowledge base holds the facts about the subject area of the expert system.
earns: "4(b) Four from: • Knowledge base holds facts (1)" Mark scheme 9626/11, November 2025, p. 5
2. These facts have been gathered from a human expert in that subject.
earns: "Knowledge base holds facts (1) gathered from an expert (1)" Mark scheme 9626/11, November 2025, p. 5
3. When a user enters a query, the inference engine interrogates the knowledge base.
earns: "It is interrogated by the inference engine (1st)" Mark scheme 9626/11, November 2025, p. 5
4. The inference engine only uses the rules that apply to that query.
earns: "It is interrogated by the inference engine (1st) – Using relevant rules/rules that apply (1)" Mark scheme 9626/11, November 2025, p. 5

A weak answer of the kind the examiners describe (written for this guide; likely 1-2 of 4):

The knowledge base stores information and rules. The expert system uses it to make decisions.

'Stores rules' matches a mark-scheme point, and 'stores information' may get the 'holds facts' mark. It never says the facts come from an expert, that the inference engine interrogates the knowledge base, or that only the relevant rules are used.

"For example, whilst many candidates stated that the knowledge base holds facts, very few stated that these were sourced from experts. Similarly, candidates were aware that rules are used to interrogate the knowledge base, but very few stated that only the relevant rules were used." Examiner report, November 2025, Paper 9626/11, p. 2

This is a 'describe how' question, so the marks are for what happens, not why. Two of the four marks are expansions of an earlier point (gathered from an expert; only relevant rules), so precise detail earns as much as new points.

WORKED EXAMPLE

Q4(a) [4 marks] “The directors of a company wish to create a database of customers. They are not sure whether they should ask a computer programmer to produce custom written software or they should purchase off-the-shelf software. (a) Describe the features of custom written software.” Question paper 9626/13, June 2022, p. 4

A full-mark answer, built from the mark scheme (not a real candidate's):

1. Custom written software is specially developed for one specific company, here the directors' company.
earns: “Software that is specially developed for a specific company/business” Mark scheme 9626/13, June 2022, p. 4
2. It is made to accommodate that company's particular preferences and needs, such as the exact customer details the directors want to store.
earns: “It is made to accommodate that customer's particular preferences and needs” Mark scheme 9626/13, June 2022, p. 4
3. Support is obtained directly from the programmers who created it.
earns: “Support is obtained directly from the creators” Mark scheme 9626/13, June 2022, p. 4
4. The development costs have to be paid by the company that commissions it.
earns: “Development costs have to be paid for by client for whom its written/ commissioning company/business” Mark scheme 9626/13, June 2022, p. 4

A weak answer of the kind the examiners describe (written for this guide; likely 0 of 4):

Custom written software is more expensive. It is better than off-the-shelf software.

'More expensive' is a vague comparison with no detail, and 'better' is an opinion, not a feature. The cost point only earns a mark when it says who pays the development costs.

“Some weaker responses contained responses such as ‘more expensive’ which is not enough at this level and statements such as this should always be expanded.” Examiner report, June 2022, Paper 9626/13, p. 8

A 'describe the features' question wants four separate facts about what the thing is. No reasons are needed, but each fact must be specific enough to match a mark-scheme point.

How much evidence: Evidence is plentiful: Describe is one of the most common command words on Papers 1 and 3. The mark schemes do not define the (1st) label; the reading here (a first point that a later linked point expands) comes from how it is laid out and from the examiner report comments about expanding fewer points.

Explain

Explain — Cambridge's definition: "set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence" Syllabus 2025–2027, p. 66

Make a point, then add a reason for it with 'because', 'so' or 'which means', and repeat until you have covered the marks.

HOW THE MARKS ARE GIVEN

1. Some Explain mark schemes mark a point '(1st)' and the expansion under it '(1)'. The (1st) mark is for your statement. The (1) mark is only for the reason or detail you add to that statement. Of the question parts we counted, 38 of the 176 Explain parts from 2022 on (to March 2026) use (1st) in the mark scheme.

"The sampling rate can affect the quality of audio files/how the audio sounds (1st) – because it captures sounds at discrete points (and not the whole sound) (1)" Mark scheme 9626/12, March 2024, p. 9

"To define/specify the timing of the movement (1st)" Mark scheme 9626/32, November 2023, p. 4

"Access control can be based on attribute of user within company (1st) – so that they can only access data appropriate for their job/task/role (1)" Mark scheme 9626/32, March 2024, p. 7

2. A point with no reason can still get its one mark, but no more. If every point you write is only a statement, the mark scheme can cap you below full marks.

"A less detailed answer, such as 'they can learn at their own pace' can be given credit as it is a correct statement of the effect, but it lacks the detail to gain any further credit because it does not add an expansion" Examiner report, June 2024, Paper 9626/32, p. 23

"MAX two marks if answers are all only first marks." Mark scheme 9626/13, November 2023, p. 4

3. Where the mark scheme links an expansion to a first point, the expansion mark depends on it. In one encryption question, the two further marks were only available to answers that first said the data is scrambled. In a 2-mark question, one full point and its reason can be the whole answer.

"A relatively easy first mark was given for any answer that stated that the data is scrambled. Many candidates achieved this mark. Two further marks were then dependant on this answer." Examiner report, June 2024, Paper 9626/11, p. 4

"Max two for one complete explanation" Mark scheme 9626/12, March 2024, p. 9

HOW TO WRITE IT

1. Write explanations in full sentences, not bullet points of a few words. A sentence lets you put the point and its reason together.

"Candidates should be encouraged to write full sentences and discouraged from writing bulleted, short statements in their responses. Descriptions or explanations can only be conveyed in full sentences." Examiner report, November 2022, Paper 9626/31, p. 12

WHAT GOES WRONG

1. Do not just describe what something is. Explain asks 'why' or 'how'. Examiners say many answers gave descriptions and lost marks.

"Most responses gave descriptive answers rather than explanations." Examiner report, November 2023, Paper 9626/12, p. 6

"Responses that described what each server was did not answer the question." Examiner report, November 2022, Paper 9626/31, p. 12

"there is usually little or no credit given for describing what the point or process is since this does not answer the question 'how'" Examiner report, November 2022, Paper 9626/31, p. 12

2. Leaving out the reason is a common mistake. Short, vague phrases like 'store web sites' or 'makes the code more readable' do not get full marks.

"A common mistake was to omit a reason or expansion to the response – the question asked for explanations" Examiner report, November 2023, Paper 9626/32, p. 16

"Vague responses such as 'store web sites' were not sufficient to be awarded marks." Examiner report, November 2022, Paper 9626/31, p. 12

"a significant number only stated that comment make the code more readable so did not gain the full marks." Examiner report, March 2024, Paper 9626/32, p. 8

3. Use exact technical words. In one encryption question, about as many answers said encrypted data 'cannot be read' as said it 'cannot be understood'. Only 'cannot be understood' got the mark.

"Many candidates made the point that encrypted data can not be understood, but an approximately equal number stated that encrypted data could not be read, and were not awarded marks." Examiner report, June 2024, Paper 9626/11, p. 4

TIPS

1. An explanation has two parts. First you make a statement. Then you give a reason, a purpose or more detail about that statement.

"An explanation requires a statement or description followed by a reason or purpose or further expansion of that statement" Examiner report, June 2024, Paper 9626/32, p. 23

"Explanations require a statement and an expansion, which is often a reason but can be additional facts, to fully answer the question." Examiner report, November 2022, Paper 9626/31, p. 14

"An explanation is statement of fact with a reason or expansion giving its purpose." Examiner report, March 2024, Paper 9626/32, p. 8

2. Join the statement to its reason with a linking word such as 'so', 'so that', 'because' or 'which means'. The mark schemes and examiner reports use these words to show where the reason starts. Write the link word, so the examiner can see where your reason is.

"Can be adapted (automatically) in response to changing conditions/change of staff (1st) – so that new employees can have access (1)" Mark scheme 9626/32, March 2024, p. 7

"The lower the sample rate, the smaller the file size (1st) – because there is less data stored (1)" Mark scheme 9626/12, March 2024, p. 9

"A good answer would have stated that there is no central governing authority which means that there is no oversight or regulation" Examiner report, November 2022, Paper 9626/31, p. 14

"stating the task 'to check that the project has ended' should have been followed by a reason such as 'so that all stakeholders and clients can agree that no further additions or changes are necessary'" Examiner report, June 2025, Paper 9626/33, p. 25

3. If the question has a scenario, use it. Your reasons should fit the situation in the question, not just be general facts.

"As with all scenario-based questions, candidates are advised to consider the significance of this scenario." Examiner report, November 2023, Paper 9626/13, p. 9

"The full range of marks is only available to candidates for answers referring to the scenario in the questions." Examiner report, November 2022, Paper 9626/31, p. 12

WORKED EXAMPLE

Q3(c) [3 marks] “Mainframe computers were developed from the need for computers to process data in tasks such as carrying out a census. (c) Explain why mainframe computers are used to process the data from a census.” Question paper 9626/13, November 2023, p. 3

A full-mark answer, built from the mark scheme (not a real candidate's):

1. A census collects a very large amount of data that has to be processed,
earns: “The amount of data that has to be processed is very large (1st)” Mark scheme 9626/13, November 2023, p. 4
2. as countries tend to have a lot of people and each person adds more data.
earns: “as countries tend to have a lot of people (1)” Mark scheme 9626/13, November 2023, p. 4
3. All of this data also needs to be processed in a short period of time, so a mainframe with a high processing speed is used.
earns: “All data needs to be processed in a short period of time (1st)” Mark scheme 9626/13, November 2023, p. 4

A weak answer of the kind the examiners describe (written for this guide; likely 2 of 3):

There is a lot of data in a census. Mainframes are very reliable.

Both sentences are only first marks with no reason after them. The mark scheme says answers that are all first marks get a maximum of two, and the examiners said most answers stopped at 'a large amount of data'.

“Typically, however, responses tended to focus on that fact that there is usually a large amount of data.” Examiner report, November 2023, Paper 9626/13, p. 9

Adding 'as countries tend to have a lot of people' turns a statement into an explanation and unlocks the third mark. Think about why the scenario (a census) needs a mainframe.

WORKED EXAMPLE

Q10(a) [3 marks] “Explain how each of these methods keeps data safe from unauthorised users. (a) encryption” Question paper 9626/11, June 2024, p. 11

A full-mark answer, built from the mark scheme (not a real candidate's):

1. Encryption scrambles the data,
earns: “Data is scrambled (1st)” Mark scheme 9626/11, June 2024, p. 9
2. so that it cannot be understood by anyone who intercepts it,
earns: “so it is not understandable (1)” Mark scheme 9626/11, June 2024, p. 9
3. which means the data is of no use to a hacker or any other third party.
earns: “of no use to a hacker/third party (1)” Mark scheme 9626/11, June 2024, p. 9

A weak answer of the kind the examiners describe (written for this guide; likely 0 of 3):

Encryption makes the data unreadable so hackers cannot read it.

It does not say the data is scrambled, so the first mark is missing and the two expansion marks cannot follow. 'Cannot be read' is the wrong idea; the examiners wanted 'cannot be understood'.

“Rather than making data unreadable, encryption puts data into a state whereby it can not be understood.” Examiner report, June 2024, Paper 9626/11, p. 4

One point and two linked reasons can earn all three marks here. The first statement is the gateway; each linked reason after it adds a mark.

How much evidence: Explain is common (176 parts from 2022 to March 2026), so the evidence is broad. But only 38 of those mark schemes use the explicit (1st)/(1) labels; the rest simply list points worth one mark each. The cap for answers that are 'all only first marks' is written out in November 2023 mark scheme (Paper 13) Q3(c). No other mark scheme found prints that cap, so treat it as a sign of how markers think, not a rule for every Explain question.

Discuss

Discuss — Cambridge's definition: "write about issue(s) or topic(s) in depth in a structured way" Syllabus 2025–2027, p. 66

Write full sentences in paragraphs that give benefits and drawbacks of the thing in the question (aim for at least two of each), and expand every point with a 'because' or 'so'.

HOW THE MARKS ARE GIVEN

1. Marks go to points from a list. A point is often worth 1 mark, and a second mark comes when you expand it: the mark scheme shows the point as (1st) and the expansion as (1). A bare point with no expansion leaves marks behind.

"Direct data source/direct data is reliable/trusted (1st) – because we collected it (1)" Mark scheme 9626/12, June 2024, p. 7

"Orders are/can be processed straight away (1st) • so process of filling/completing orders can start sooner (than for batch processing) (1)" Mark scheme 9626/11, November 2023, p. 9

"Must have expansions or comparisons to be a proper discussion" Mark scheme 9626/12, March 2022, p. 4

2. Each side often has a ceiling. In the 2022-2025 mark schemes we checked (77 Discuss parts), about 36 print a separate MAX for each side; for 8-mark questions it is usually MAX 6 (sometimes 5, once 7). On those questions an answer with only benefits cannot reach full marks.

"This method gathers direct data, which is more up to date (1) Benefits max 6:" Mark scheme 9626/12, November 2024, p. 9

"Disadvantages MAX five from" Mark scheme 9626/12, June 2023, p. 7

"Content areas could include: Max SIX from: Benefits: e.g.:" Mark scheme 9626/31, June 2025, p. 12

3. One-sided answers are capped. 23 of the 77 mark schemes we checked (all 2022-2023) say you need at least two of each side (one of each on some 4- and 6-mark questions) for full marks, and 4 more (November 2023) cap an all-benefits or all-drawbacks answer. The 2023 and 2025 examiner reports say the same: no full marks without both sides.

"Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks" Mark scheme 9626/12, March 2022, p. 4

"Answers that do not provide both do not access the highest marks because a proper discussion should look at both sides of an issue." Examiner report, June 2023, Paper 9626/32, p. 22

"Also, where both benefits and drawbacks are required by the question, answers that address only benefits or only drawbacks cannot achieve full marks." Examiner report, June 2025, Paper 9626/32, p. 22

4. Sometimes there is a mark for a short opening definition or description of the thing in the question. We found it in 18 of the 77 mark schemes, usually capped at one mark for each thing described (two marks once, in June 2025 Paper 31 Q7). So one clear opening sentence can help, but a long introduction cannot earn more.

"One mark from: • Definition of encryption (1) Max 6 from: Benefits:" Mark scheme 9626/12, June 2025, p. 11

"Max one for: • Definition of computer simulation (1) Max six from: Benefits" Mark scheme 9626/11, November 2025, p. 9

"Eight from: Simple statement of the difference between the two (1): e.g." Mark scheme 9626/12, June 2024, p. 7

5. No 2022-2025 Discuss mark scheme we checked has a mark for a conclusion or your opinion. The current syllabus meaning of Discuss does not ask for one. Older reports (2019) used an older wording that said Discuss often needs a conclusion; that is no longer how it is defined. A one-line conclusion does no harm, but put your time into expanded points.

"Discuss write about issue(s) or topic(s) in depth in a structured way" Syllabus 2025–2027, p. 66

"Discuss questions require candidates to give the important arguments for and against, and often require a conclusion." Examiner report, November 2019, Paper 9626/12, p. 4

HOW TO WRITE IT

1. Bullet points cost marks. Every 2022 and 2023 Discuss mark scheme (and 6 of the 17 in 2024) caps a bulleted list: max 6 of 8, max 4 or 5 of 6, max 3 of 4. We counted 48 of the 77 mark schemes with this cap.

"Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion" Mark scheme 9626/12, March 2022, p. 4

"Must be at least one of each for full marks. Max 4 marks if bullets/list of points." Mark scheme 9626/32, March 2023, p. 11

"Must have at least one of each to gain full marks Max. three marks if bullets/list of points" Mark scheme 9626/12, November 2022, p. 8

2. The 2025 mark schemes no longer print the bullet cap, but the examiners still say lists and tables of statements cannot reach most of the marks. This includes two separate lists (benefits, then drawbacks) made of single words or very short sentences.

"So learners who produced bulleted lists or tables of statements were unable to access most of the marks." Examiner report, March 2025, Paper 9626/32, p. 10

"The command word 'discuss' requires candidates to provide a structured answer, so a bulleted list of brief statements is not properly answering the question and does not gain access to the full range of marks." Examiner report, June 2024, Paper 9626/32, p. 23

"This comment also applies to answers that have separate lists of benefits and drawbacks, but the lists consist of single word or very brief sentences." Examiner report, June 2025, Paper 9626/32, p. 22

3. Write in full sentences and continuous prose. A good shape is one paragraph of benefits and one of drawbacks, with each point followed by its reason or effect in the same sentence.

"Candidates should be encouraged to write answers in full sentences." Examiner report, June 2025, Paper 9626/31, p. 17

"Questions that ask candidates to discuss the benefits and drawbacks of something require responses to give both sides of the argument and should be in continuous prose rather than bullet points." Examiner report, March 2022, Paper 9626/12, p. 1

"The strongest answers were written in full prose style and included points which were expanded upon" Examiner report, March 2022, Paper 9626/12, p. 2

WHAT GOES WRONG

1. Making a point and stopping. Examiners say most students give valid points but few expand them, and two 2022 mark schemes cap answers with no expansion at 6 of 8.

"Most responses gained credit for making valid points, but few went on to discuss the points they made by expanding on them." Examiner report, March 2022, Paper 9626/12, p. 2

"Max. six marks if bullets/list of points Max six marks if no attempt at expansion" Mark scheme 9626/12, November 2022, p. 10

2. Vague cost words. Writing that something is 'expensive' or 'costs more' was too vague. Say what the money is spent on (for example travelling to interviews or printing questionnaires).

"Where candidates stated that collecting direct data was expensive, this was considered too vague, but if candidates identified an area on which money would have to be spent, which would not otherwise be spent, this was awarded." Examiner report, June 2024, Paper 9626/12, p. 7

"Many responses mentioned costs without going into sufficient detail to merit credit." Examiner report, March 2022, Paper 9626/12, p. 2

3. Writing the same point twice, once as a benefit and again reversed as a drawback. It earns only once and wastes time.

"Some candidates effectively wrote what was good about direct data compared to indirect data and then repeated the list, but in reverse structure, so the positive aspects of one became the negative points of the other." Examiner report, June 2024, Paper 9626/12, p. 7

"(Converse answer may also be given as an advantage – Award ONCE only)" Mark scheme 9626/12, June 2023, p. 7

4. Describing the topic instead of discussing it in the question's setting. Describing types of technology, or the stages of a process, or writing about the wrong context, gained little or nothing. Tie each benefit and drawback to the people or organisation named in the question.

"Descriptions of different types of TEL, statements about benefits or drawbacks gained little or no credit without further comment on how these can affect the achievement of students." Examiner report, March 2025, Paper 9626/32, p. 10

"there were many who made mistakes such as describing the stages of the data mining process or confusing data mining with another topic altogether" Examiner report, June 2025, Paper 9626/31, p. 19

"Answers should have focused on how banking organisations make use of data mining." Examiner report, June 2025, Paper 9626/32, p. 23

WORKED EXAMPLE

Q7 [8 marks] "Data can be collected by using a direct or an indirect data source. Discuss the advantages and disadvantages of using direct data compared to indirect data." Question paper 9626/12, June 2024, p. 8

A full-mark answer, built from the mark scheme (not a real candidate's):

1. Direct data is collected first-hand by the person who will use it (a primary source), while indirect data was collected by someone else for another purpose (a secondary source).

earns: "Simple statement of the difference between the two (1): e.g." Mark scheme 9626/12, June 2024, p. 7

2. One advantage of direct data is that it is more reliable and trusted, because we collected it ourselves and know exactly how it was gathered.

earns: "Direct data source/direct data is reliable/trusted (1st) – because we collected it (1)" Mark scheme 9626/12, June 2024, p. 7

3. Also, only specific, relevant data is collected, so time is not wasted sorting through data we do not need.

earns: "ONLY Specific/relevant data is collected (1)" Mark scheme 9626/12, June 2024, p. 7

4. Direct data is also more up to date than indirect data, which may have been collected years ago.

earns: "Direct data is (more) up-to-date (1) Disadvantages MAX SIX" Mark scheme 9626/12, June 2024, p. 7

5. However, a disadvantage is that the sample size may be small, whereas indirect data sources tend to be much larger sets of data.

earns: "The sample size may be small (1st) – whereas indirect data sources tend to be larger sets of data (1)" Mark scheme 9626/12, June 2024, p. 7

6. Collecting direct data also costs money that would not otherwise be spent, such as travelling to interview people and printing questionnaires.

earns: "Collecting direct data involves the expense of travelling/photocopying/printing questionnaires//any expense to do with a direct data method (1)" Mark scheme 9626/12, June 2024, p. 7

A weak answer of the kind the examiners describe (written for this guide; likely about 2 of 8 (estimate)):

Advantages: direct data is cheaper, relevant, up to date. Disadvantages: indirect data is not relevant and old, direct data is expensive.

It is a list, not a discussion. 'Expensive' does not say what the money is spent on. The disadvantages just repeat the advantages in reverse, so they earn nothing new.

"Some candidates effectively wrote what was good about direct data compared to indirect data and then repeated the list, but in reverse structure, so the positive aspects of one became the negative points of the other." Examiner report, June 2024, Paper 9626/12, p. 7

Six sentences earn eight marks because two of them carry a point plus its expansion. The examiner report says this question was really about what was good and bad about direct data, with indirect data only as the comparison point, so do not write each point twice in reverse.

WORKED EXAMPLE

Q7 [8 marks] “Discuss the benefits and drawbacks of using data mining in businesses.” Question paper 9626/31, June 2025, p. 8

A full-mark answer, built from the mark scheme (not a real candidate’s):

- 1. Data mining is the use of software tools to analyse very large data sets.
earns: “data mining (is a set of techniques and tools) used to from work on / analyse large data sets” Mark scheme 9626/31, June 2025, p. 12
- 2. Its aim is to discover trends, patterns and relationships in the data that people would not spot by hand.
earns: “to discover trends / patterns / to find relationships” Mark scheme 9626/31, June 2025, p. 12
- 3. A business can use data mining to improve its sales and profits, because it can target its advertising at the customers or demographic groups most likely to buy.
earns: “used to improve sales / profits (1st) by targeting customers / demographics (1)” Mark scheme 9626/31, June 2025, p. 12
- 4. The information it extracts also gives managers a sound basis for their decisions, rather than guesswork.
earns: “provides a sound basis for decision-making” Mark scheme 9626/31, June 2025, p. 12
- 5. On the other hand, the extracted information may not be accurate, because it depends on how good the source data set is.
earns: “extracted information may not be accurate (1st) depends on source of data set (1)” Mark scheme 9626/31, June 2025, p. 12
- 6. The business must also protect the privacy of the individuals in the data and follow data protection laws, which adds work and risk.
earns: “must ensure privacy of individuals / conform to data protection regulations” Mark scheme 9626/31, June 2025, p. 12

A weak answer of the kind the examiners describe (written for this guide; likely low (estimate: 2-3 of 8)):

A two-column table: Benefits | Drawbacks; trends | expensive; profit | privacy; decisions | storage.

Single words in a table make points but give no reasons or effects, so there is no discussion and no expansion marks.

“Furthermore, if these answers are shown in a sketched table, they do not properly answer the question as there is no discussion.” Examiner report, June 2025, Paper 9626/31, p. 17

This 2025 mark scheme allowed up to two marks for saying what data mining is, then MAX SIX per side. A two-sentence definition, plus two benefits and two drawbacks (one of each expanded with its reason), reaches eight.

How much evidence: Not thin: Discuss appears in 80 question parts from 2022 onward (77 with mark schemes available). The counts above are our own counts from the mark-scheme text and could be off by one or two.

Evaluate (and Assess)

Evaluate — Cambridge's definition: "judge or calculate the quality, importance, amount, or value of something" Syllabus 2025–2027, p. 66

Assess — Cambridge's definition: "make an informed judgement" Syllabus 2025–2027, p. 66

Give points for AND against, add a consequence to each, say how important it is in this scenario, and finish with a reasoned overall judgement.

HOW THE MARKS ARE GIVEN

1. You must write about both sides. Some 2022-2023 mark schemes cap answers that are all for or all against at 6 of 8 (3 of the 24 Evaluate mark schemes from 2022 on), and two say you need at least two points on each side for full marks.
 - "Max 6 if all for or all against. Max 6 marks if bullets/list of points." Mark scheme 9626/32, November 2023, p. 10
 - "Must be at least two of each for full marks. Max 6 marks if bullets/list of points." Mark scheme 9626/32, March 2023, p. 7
 - "Must be at least 2 strengths and at least 2 weaknesses for full marks." Mark scheme 9626/32, November 2022, p. 6
2. Recent mark schemes give one mark for a conclusion. We checked the Evaluate parts from 2022 onward: 8 of the 24 with a mark scheme have a conclusion mark, and all 8 are from November 2024 or later (8 of the 9 mark schemes from that date on). End with a short overall judgement that pulls your points together.
 - "Valid conclusion drawing together/based on arguments presented in the answer (1)" Mark scheme 9626/13, June 2025, p. 9
 - "Max one from: Conclusion: • One mark for a reasoned conclusion / overall judgement" Mark scheme 9626/32, November 2025, p. 10
 - "One mark is available for a valid conclusion / judgement (must have considered both pros and cons)." Mark scheme 9626/32, November 2024, p. 15
 - "Such discussions should end with a conclusion that revisits and brings together the points made." Examiner report, November 2024, Paper 9626/12, p. 4
3. Good answers take each point through three steps: a fact, a consequence of that fact, and a judgement about how much it matters. Some mark schemes give a first mark for the point and further marks for the expansion, and limit how many marks plain points can earn.
 - "Good answers gave a description of how compression was used or of the effects of the compression and then added a consequence followed by a judgment as to the desirability or further effect of the compression." Examiner report, June 2025, Paper 9626/33, p. 26
 - "Eight marks from: MAX four for 1st marks only Evaluations." Mark scheme 9626/13, June 2025, p. 9
 - "Must have expansions to be a proper evaluation" Mark scheme 9626/12, March 2022, p. 8
4. Some mark schemes give one mark for a short definition or description of the thing being evaluated. Keep it to one sentence, then move on to judging it.
 - "One mark for a description of encryption (e.g. data scrambled/converted to cyphertext)" Mark scheme 9626/12, March 2024, p. 9
 - "Valid definition / description of video-conferencing in education" Mark scheme 9626/31, June 2025, p. 15

HOW TO WRITE IT

1. Write in full sentences and paragraphs, not bullet points or a table. 9 of the 24 Evaluate mark schemes from 2022 on (all 2022-2024) cap a bulleted list, usually at 6 of 8 marks (5 of 6 on 6-mark questions).

"A detailed answer is needed, which cannot be achieved by writing bullet points." Examiner report, March 2023, Paper 9626/12, p. 1

"Must be a proper evaluation for full marks. Max 6 marks if bullets/list of points." Mark scheme 9626/32, March 2022, p. 6

"Max. five marks if all three methods not covered Max. five marks if bullets/list of points" Mark scheme 9626/12, March 2023, p. 8

"Answers with bulleted, short statements for free response questions, especially those requiring analyses, discussions and evaluations do not adequately answer the question." Examiner report, June 2025, Paper 9626/31, p. 17

WHAT GOES WRONG

1. Leaving out a conclusion can cost marks. In the March 2024 report (Paper 12, Q9), examiners said candidates' failure to reach a realistic conclusion reduced their marks.

"candidates largely failed to come to any realistic conclusion about encryption, which reduced the marks scored on the question." Examiner report, March 2024, Paper 9626/12, p. 3

2. The most common failure is describing how the thing works instead of judging it. Knowing the topic is not enough; you have to say whether each feature helps or hinders.

"Candidates seemed to have some understanding of each process, but preferred to describe each of these than actually evaluate their efficacy." Examiner report, March 2023, Paper 9626/12, p. 3

"In some cases, candidates described how control technologies worked. Typically, this involved an in-depth discussion of sensors." Examiner report, November 2024, Paper 9626/12, p. 3

"candidates were expected to make factual statements about project management software (PMS) and then expand on these with comments about whether these facts helped or hindered the project." Examiner report, November 2023, Paper 9626/33, p. 21

3. One-sided answers lose marks. Both sides must be there, but they do not need to be exactly equal: more points for than against was accepted.

"some candidates seemed not to understand the importance of 'evaluate' as a command word, and so gave a one-sided answer." Examiner report, June 2024, Paper 9626/11, p. 1

"Candidates tended to concentrate on the positive implications, rather than both the positive and the negative, and this did restrict the overall marks scored for this question in a few cases." Examiner report, June 2025, Paper 9626/12, p. 8

"Typically, candidates gave more answers in support than against, but this was acceptable." Examiner report, June 2025, Paper 9626/13, p. 12

TIPS

1. Judge the thing in the scenario you were given, not in general. Answers about spreadsheets that never mentioned financial modelling did badly.

"To gain full credit, candidates must produce a reasoned evaluation that weighs up the advantages and disadvantages for the given scenario/context." Examiner report, March 2023, Paper 9626/12, p. 1

"Whilst some lenience was allowed, candidates who failed to make any mention of financial modelling struggled to score well on this question." Examiner report, June 2025, Paper 9626/13, p. 12

2. Evaluate only what the question names. You do not need to compare it with other methods unless the question asks you to.

"It was not necessary to compare pilot with other possible methods of implementation because the question asked candidates to 'evaluate pilot implementation' and made no mention of the other methods." Examiner report, November 2023, Paper 9626/32, p. 17

3. Evaluate is not the same as Discuss. Discuss asks you to write in depth in a structured way; Evaluate asks you to judge importance or value. Your judgement can be on each point or on the whole answer. We checked the 77 Discuss mark schemes from 2022 onward: none has a conclusion mark, while recent Evaluate mark schemes do.

"Discuss write about issue(s) or topic(s) in depth in a structured way Evaluate judge or calculate the quality, importance, amount, or value of something" Syllabus 2025–2027, p. 66

"The command word 'evaluate' requires a judgement of the importance of the points being made about the method of prototyping." Examiner report, November 2024, Paper 9626/32, p. 17

"In questions where an evaluation is, or evaluations are, required, the judgement can be for individual points or for an overall judgement on the whole content of the answer." Examiner report, November 2024, Paper 9626/33, p. 21

WORKED EXAMPLE

Q10 [6 marks] "A person uses a dialogue interface to control multiple devices in their home. Evaluate the use of a dialogue interface in this context." Question paper 9626/12, June 2025, p. 9

A dialogue interface is controlled by speaking to it.

A full-mark answer, built from the mark scheme (not a real candidate's):

1. Because the person controls the devices by speaking, they do not need to physically touch any switch or screen.

earns: "No need to physically interact (may exemplify) (1st)" Mark scheme 9626/12, June 2025, p. 9

2. This is a real benefit for a disabled person, who can switch on lights or heating around the home without moving to them.

earns: "More accessible to disabled users (may refer to device itself or devices around the home)

(1)" Mark scheme 9626/12, June 2025, p. 9

3. It also means the person can carry on with other tasks, such as cooking, while giving a command.

earns: "Can undertake other tasks at the same time (1)" Mark scheme 9626/12, June 2025, p. 9

4. However, the interface relies on sound, so loud music or a television in the room may stop it understanding the command.

earns: "External noise may result in the dialogue interface not understanding the command

(1)" Mark scheme 9626/12, June 2025, p. 9

5. It may also mishear what is said and operate the wrong device, which reduces how far the person can rely on it.

earns: "Dialogue interface may misinterpret what is said (1)" Mark scheme 9626/12, June 2025, p. 9

6. Overall, a dialogue interface is very useful in a home, especially for someone with limited movement, as long as the rooms are quiet enough for commands to be understood.

earns: "One mark for a valid conclusion (1)" Mark scheme 9626/12, June 2025, p. 9

A weak answer of the kind the examiners describe (written for this guide; likely about 2 of 6):

Dialogue interfaces are good for disabled people. You control things with your voice. It is hands-free and easy to use.

Every point is positive, nothing is expanded into a consequence, and there is no conclusion, so most of the marks for drawbacks and for the judgement cannot be reached.

"Candidates tended to concentrate on the positive implications, rather than both the positive and the negative, and this did restrict the overall marks scored for this question in a few cases." Examiner report, June 2025, Paper 9626/12, p. 8

Three sentences for, two against, one conclusion. The conclusion mark is easy to get if you remember to write one.

WORKED EXAMPLE

Q10 [8 marks] “A company is developing a data input screen for entering customer details into a database. Evaluate the use of evolutionary prototyping for developing the data input screen.” Question paper 9626/32, November 2024, p. 11

A full-mark answer, built from the mark scheme (not a real candidate's):

1. In evolutionary prototyping, the data input screen is continually refined and rebuilt until it is finished.
earns: “The system / screen is continually refined and rebuilt (when developed using evolutionary prototyping)” Mark scheme 9626/32, November 2024, p. 15
2. The staff who will use the screen work with every version of it as it is developed.
earns: “The user continuously engages with the system (1)” Mark scheme 9626/32, November 2024, p. 15
3. As a result, the finished screen is more likely to include the customer fields the staff actually need, which is the most important benefit for a data input screen.
earns: “so the new system / screen is more likely to meet the user requirements (1)” Mark scheme 9626/32, November 2024, p. 15
4. The screen can also be changed at any iteration, for example if the company decides to record extra customer details.
earns: “Screen / product can be changed at any stage / iteration (1)” Mark scheme 9626/32, November 2024, p. 15
5. On the other hand, the constant meetings between developers and users can be costly.
earns: “Continual need to consult / interact between developers / client / end-users can be costly (1)” Mark scheme 9626/32, November 2024, p. 15
6. This raises the overall cost of development, which is a serious drawback when only one screen is being built.
earns: “which raises overall cost of development (1)” Mark scheme 9626/32, November 2024, p. 15
7. Developers may also find it hard to know when to stop tweaking the screen, so development may go on too long.
earns: “Developers may have difficulty knowing when it is necessary to stop ‘tweaking’ / developing the system (1)” Mark scheme 9626/32, November 2024, p. 15
8. Overall, evolutionary prototyping suits this screen because user feedback matters most for data entry, provided the company sets a clear end point to control the cost.
earns: “One mark is available for a valid conclusion / judgement (must have considered both pros and cons).” Mark scheme 9626/32, November 2024, p. 15

A weak answer of the kind the examiners describe (written for this guide; likely probably no more than 3 of 8):

Evolutionary prototyping involves the users. The screen can be changed. It can be expensive. It takes a long time.

These are bare statements. None is followed by a consequence or a judgement, and there is no conclusion.

“Weaker answers confused the methods of prototyping, made statements with no consequences, expansion, or judgments.” Examiner report, November 2024, Paper 9626/32, p. 17

The same facts appear in both answers. The strong answer earns more because each fact is followed by what it leads to and how much it matters, and it ends with a verdict.

How much evidence: Evaluate is well covered: 25 question parts from 2022 onward (114 in all years). Assess is in the syllabus command-word list, but no question part uses Assess as its command word, and a search of all question paper pages found no question starting with 'Assess'. There are no Assess mark schemes or examiner comments, so this report gives no separate advice for Assess beyond its syllabus definition ('make an informed judgement'). The conclusion-mark count rests on a text search of the mark schemes for 'conclusion' or 'overall judgement'.

Analyse

Analyse — Cambridge's definition: "examine in detail to show meaning, identify elements and the relationship between them" Syllabus 2025–2027, p. 66

Break the topic into its parts, and for each part say what it leads to in the situation in the question: point, then 'so', then the consequence.

HOW THE MARKS ARE GIVEN

1. Marks come from making a point and then expanding it into what it causes or means. Some mark schemes say an answer without expansions is not a proper analysis, and two 2022 ones cap it at 6 of 8. We checked the 18 Analyse mark schemes from 2022 onward: none gives a mark for a conclusion.

"Max six marks if bullets/list of points Max six marks if no attempt at expansion" Mark scheme 9626/13, November 2022, p. 6

"Must be a proper analysis to gain full marks Max. six marks if bullets/list of points Must have expansions to be a proper analysis" Mark scheme 9626/11, June 2022, p. 4

"candidates should produce answers that make a point and then expand on the point." Examiner report, June 2023, Paper 9626/32, p. 22

2. Show the relationships between the parts: one thing leads to another, which leads to a further effect. Examiners and mark schemes reward these chains of cause and effect.

"Good answers should have made references to the need for greater bandwidth, for example, so IT infrastructure has to be upgraded which increases costs taking money from other needs" Examiner report, June 2023, Paper 9626/31, p. 18

"Increased digital awareness leads to increased/better infrastructure// more investment in IT (accept personal equivalences) (1) which then leads to increased digital awareness (1)" Mark scheme 9626/12, March 2024, p. 10

"because at each compression/decompression more image information is lost (1)" Mark scheme 9626/32, November 2023, p. 5

3. A short description or definition at the start can earn one or two marks, but no more. Description only earns marks when you use it to explain the effect or the reason.

"Max two from: Resolution is measured/described as pixels per inch/ppi" Mark scheme 9626/32, March 2022, p. 5

"Credit was given for a brief description of UHD television but the command word (see. p. 45 of the syllabus) analyse requires candidates to 'examine in detail to show meaning, identify elements and the relationship between them' so this question demanded more." Examiner report, June 2023, Paper 9626/31, p. 18

"Descriptions of vector images were not given credit unless these descriptions were used to explain why vector graphics are, or are not, used on websites." Examiner report, March 2025, Paper 9626/32, p. 12

HOW TO WRITE IT

1. Write in full sentences. 10 of the 18 Analyse mark schemes from 2022 on (all 2022-2023) cap bullet points: 6 of 8, 5 of 7, or 4 or 5 of 6.

"questions that ask candidates to 'analyse' or 'discuss' require detailed responses that include a proper analysis or discussion, rather than short or bullet-pointed responses." Examiner report, March 2023, Paper 9626/12, p. 1

"Max 4 marks if bullets/list of points/characteristics." Mark scheme 9626/31, June 2023, p. 8

"Max. six marks if bullets/list of points Max six if no expansions" Mark scheme 9626/11, November 2022, p. 6

WHAT GOES WRONG

1. Writing about what the thing is or how it works, instead of the effect the question asks about. If the question asks about risks to an economy or effects on an image, every point must reach that.

"This question was not about digital currencies and what they are or how they work. It was about the risks to an economy that might occur as a result of the use of digital currencies." Examiner report, June 2023, Paper 9626/32, p. 22

"references to a reduction in file size without an accompanying consequence on the image itself did not answer the question." Examiner report, November 2023, Paper 9626/32, p. 16

2. Listing features. A list of points without expansion gets few marks.

"Responses often consisted of lists of the features of encryption." Examiner report, June 2022, Paper 9626/11, p. 2

"The weaker answers did not go beyond listing points so could not gain many marks." Examiner report, June 2023, Paper 9626/31, p. 19

3. Ignoring the focus of the question or being vague. Stay on the exact topic in the question, and avoid empty phrases like 'looks better'.

"Many candidates attempted to give answers that did not focus on the production of weather forecasts, or were too vague and little credit could be given." Examiner report, November 2023, Paper 9626/13, p. 10

"Questions about the digital divide require candidates to concentrate on the specific focus, otherwise answers tend to be generic, unfocused and, usually, largely irrelevant." Examiner report, March 2024, Paper 9626/12, p. 3

"Vague references to 'vector graphics make the site look better' or 'look attractive to students' did not gain credit." Examiner report, March 2025, Paper 9626/32, p. 12

TIPS

1. Cover good and bad sides where they exist. It is not always needed for full marks, but some mark schemes cap answers that cover only one side or one item.

"While it was possible to gain full credit without doing so, a full analysis should have given details of both 'for' and 'against' the use of the waterfall method in the question scenario" Examiner report, June 2024, Paper 9626/32, p. 23

"Typically, some of the stronger responses did detail the good and the bad aspects and explain them in detail to show meaning" Examiner report, June 2022, Paper 9626/32, p. 18

"Max 4 for all lossless or for all lossy" Mark scheme 9626/32, November 2023, p. 5

2. Read the whole question. If it names something and asks about 'other' ones, the named ones earn nothing.

"a significant number of a candidates included the tools mentioned in the question in their answer, which did not gain them credit." Examiner report, June 2025, Paper 9626/32, p. 20

WORKED EXAMPLE

Q3 [6 marks] “Bitmap images can be compressed when stored and used on computer systems. Analyse the effects of the two main methods of image compression on a bitmap image.” Question paper 9626/32, November 2023, p. 4

A full-mark answer, built from the mark scheme (not a real candidate’s):

1. The two main methods of image compression are lossless compression and lossy compression.
earns: “(The two methods are) lossless compression and lossy compression” Mark scheme 9626/32, November 2023, p. 5
2. Lossless compression stores all the image information exactly, because no image information is removed.
earns: “All image information is stored exactly/no image information is removed” Mark scheme 9626/32, November 2023, p. 5
3. As a result, the image can be recreated perfectly, with no difference from the original.
earns: “Image is/can be recreated perfectly/with no difference from the original” Mark scheme 9626/32, November 2023, p. 5
4. Lossy compression loses information during compression, and that information cannot be replaced.
earns: “Information is lost during compression and is not/cannot be replaced” Mark scheme 9626/32, November 2023, p. 5
5. Because of this, the image is degraded and may look blurred or blocky.
earns: “Images are degraded/reduced in quality/blurred/pixelated/blocky” Mark scheme 9626/32, November 2023, p. 5
6. The quality gets worse each time the image is compressed and saved again, so an image that is edited many times keeps losing detail.
earns: “Image quality is further degraded/reduced with repeated compression/decompression/saving and opening (1st)” Mark scheme 9626/32, November 2023, p. 5

A weak answer of the kind the examiners describe (written for this guide; likely at most 1 of 6):

Compression makes the file size smaller, so the image takes up less storage space and is quicker to send by email. Lossy makes the file smaller than lossless.

It talks about file size and sending time, not about the effect on the image, which is what the question asked.

“references to a reduction in file size without an accompanying consequence on the image itself did not answer the question.” Examiner report, November 2023, Paper 9626/32, p. 16

Both methods are covered (the mark scheme allows at most 4 for one method), and each sentence says what happens to the image. The 'because' and 'as a result' links are the analysis.

How much evidence: 19 Analyse parts from 2022 onward (18 with a matched mark scheme), mostly on Paper 3. That is enough to see the pattern, but fewer than for Evaluate or Discuss. One examiner report comment (June 2024 Q8) says both sides were not required for full marks, so 'cover both sides' is advice, not a fixed rule.

Justify

Justify — Cambridge's definition: "support a case with evidence/argument" Syllabus 2025–2027, p. 66

Argue only FOR the choice in the question: state a benefit, then give the reason it helps in this scenario, and never list drawbacks of the choice.

HOW THE MARKS ARE GIVEN

1. Most marks go to the reasons. About half the Justify mark schemes (9 of 19 from 2022 on) give a first mark for a benefit and a second for the reason or result that follows from it.

"It is the reasons that gain most credit in questions that ask for 'justify'." Examiner report, November 2024, Paper 9626/33, p. 19

"Give a positive impact (1st) • Give an extension that is a positive implication on a customer or business of the identified justification. (1)" Mark scheme 9626/13, November 2024, p. 8

"One mark per valid validation check identified (max 5) and justification given to a maximum of eight marks." Mark scheme 9626/12, March 2025, p. 7

2. Some mark schemes give one mark for a definition of the thing you are justifying. One clear sentence is enough.

"Any correct definition of direct data sources, such as:" Mark scheme 9626/11, June 2023, p. 10

"Definition of encryption (1) • Protects data/data "is safe"/improves data security (1)" Mark scheme 9626/12, November 2025, p. 8

"Definition of online processing: ONE from: • User is in direct connection with a central computer (1)" Mark scheme 9626/13, November 2024, p. 8

HOW TO WRITE IT

1. Write in full sentences. Four 8-mark Justify mark schemes (all 2023) cap a bulleted list at 6 marks.

"so less expensive media/data storage costs (1) Max 6 marks for just bullet points" Mark scheme 9626/13, November 2023, p. 9

"indirect data is not hers to sell/pass/share (1) MAX 6 marks for a bulleted list" Mark scheme 9626/11, June 2023, p. 10

"Candidates should be discouraged from writing bulleted, short statements in their responses but instead be encouraged to write answers in full sentences." Examiner report, March 2024, Paper 9626/32, p. 7

WHAT GOES WRONG

1. Justify is one-sided. Arguments against the choice earn nothing. One examiner report (June 2025) says drawbacks are not required for any question that uses Justify, and several reports describe candidates giving 'for and against' answers that did not meet the demand of the question.

"Some responses did not correctly interpret the command word of 'justify', and so gave arguments for and against. To justify an action, responses need to support an argument rather than criticise." Examiner report, November 2023, Paper 9626/12, p. 7

"Most responses also gave arguments against the decision. These answers did not meet the demand of the question and were not worthy of credit." Examiner report, November 2023, Paper 9626/13, p. 10

"Drawbacks of using mail merge were not required when answering this question, nor are they required for any question that uses the command word 'justify'." Examiner report, June 2025, Paper 9626/33, p. 25

2. Listing features or describing how something works is not justifying. Turn each feature into a reason for using it.

"Most candidates could list the features of a VPN, but few used these to justify its use." Examiner report, June 2025, Paper 9626/33, p. 26

"many candidates ignored the command word and wrote, it appeared, all they knew about lossy, and lossless, compression rather than stating facts and using these as reasons why lossy compression would be used." Examiner report, March 2024, Paper 9626/32, p. 7

"few candidates attempted to fully justify, with some candidates giving reasons against the decision, and others giving descriptions." Examiner report, November 2025, Paper 9626/11, p. 3

3. Repeating the same benefit in different words does not earn extra marks. Give a range of different reasons.

"Whilst candidates were aware of some advantages, these were often narrow and repeated from different angles in an apparent attempt to achieve more marks." Examiner report, November 2024, Paper 9626/13, p. 9

"To justify the use of an interpreter, candidates had to have a range of benefits of the use of an interpreter." Examiner report, June 2025, Paper 9626/13, p. 11

4. Vague reasons lose marks. Be technically precise about why the choice helps.

"Some of the reasons given in support were either vague or incorrect." Examiner report, November 2025, Paper 9626/12, p. 9

"In some cases, there was a lack of technical understanding, whilst in others, answers were vague and lacked clarity." Examiner report, June 2025, Paper 9626/13, p. 11

TIPS

1. You may mention a drawback of an alternative, because that still supports the choice. For example: interviews give data relevant to the study, whereas indirect sources are not specific to it.

"The command word 'justify' requires answers to support a case so drawbacks do not gain credit unless they are of alternatives." Examiner report, June 2024, Paper 9626/31, p. 20

"This justification may, for example, include why an alternative method has not been adopted." Examiner report, June 2023, Paper 9626/13, p. 11

"You are in control of the sample size//can be as big/small as needed (1st) – whereas Isabella has no control over the amount of data collected from an indirect source (1)" Mark scheme 9626/11, June 2023, p. 10

2. Tie every reason to the scenario. Full marks may only be possible when the concept and the context are combined, and a check or feature that does not fit the given data earns nothing.

"The mark scheme was structured so that full marks could only be achieved where the concept and the context were combined" Examiner report, November 2025, Paper 9626/12, p. 7

"In questions such as this, context is important. For example, a student number is unlikely to be validated with a range check, as candidates were not given a range." Examiner report, March 2025, Paper 9626/12, p. 3

"Candidates should apply their knowledge to the scenario in the question set because the full range of marks is only available to candidates for answers referring to the question scenario." Examiner report, June 2024, Paper 9626/31, p. 17

3. If the question names two things to justify, cover both. One mark scheme capped answers that dealt with only one of them at 6 of 8.

"Max six if only one of student number or examination score used." Mark scheme 9626/12, March 2025, p. 7

WORKED EXAMPLE

Q9 [4 marks] “A programmer is developing a new piece of software. The programmer has decided to write code that needs to be interpreted so that it will run. Justify the programmer’s decision.” Question paper 9626/11, November 2025, p. 7

A full-mark answer, built from the mark scheme (not a real candidate’s):

1. Code that is interpreted can be written in a high-level language, which is easier for the programmer to write and read.

earns: “Programmer can write in high level language (1)” Mark scheme 9626/11, November 2025, p. 8

2. This means the programmer does not need to code in machine code.

earns: “Programmer does not need to code in machine code (1)” Mark scheme 9626/11, November 2025, p. 8

3. The same program code will run on different computers and operating systems, as long as each has an interpreter.

earns: “The program code will run on different computers/operating systems (1)” Mark scheme 9626/11, November 2025, p. 8

4. The interpreter highlights mistakes as soon as it finds them, so the programmer can debug the new software quickly while developing it.

earns: “An interpreter will highlight mistakes as they are found (1)” Mark scheme 9626/11, November 2025, p. 8

A weak answer of the kind the examiners describe (written for this guide; likely 1 of 4):

Interpreted code is easier to debug. However, it runs more slowly than compiled code and the user needs the source code, so a compiler would be better.

Only the first sentence supports the decision. The rest argues against it, which earns nothing for Justify.

“few candidates attempted to fully justify, with some candidates giving reasons against the decision, and others giving descriptions.” Examiner report, November 2025, Paper 9626/11, p. 3

Four marks need four separate reasons in favour. Do not mention compilers unless you use them to show why the interpreter is better.

WORKED EXAMPLE

Q10 [8 marks] “The manager of the payroll department within a company has commissioned a computer programmer to write software to run the payroll. Justify this decision to use custom written software.” Question paper 9626/13, November 2023, p. 11

A full-mark answer, built from the mark scheme (not a real candidate's):

1. Custom written software is created for this exact purpose: running this company's payroll.

earns: “(This) custom written software is created for a defined/this purpose/running a payroll” Mark scheme 9626/13, November 2023, p. 9

2. So it does not need to be adapted before the payroll staff can use it.

earns: “Does not need to be adapted for use (1)” Mark scheme 9626/13, November 2023, p. 9

3. The programmer can make sure the software has all the features the manager requires, such as the company's own pay rules.

earns: “Software will have/Programmer can ensure the software will have all the features that are required (1)” Mark scheme 9626/13, November 2023, p. 9

4. It will not include unnecessary features, so payroll staff are not distracted by bloatware.

earns: “Custom written software will not have any unnecessary/distracting features/bloat ware (1)” Mark scheme 9626/13, November 2023, p. 9

5. Any settings built into the software can be changed by the programmer.

earns: “(Any) settings built into the software can be changed (1st)” Mark scheme 9626/13, November 2023, p. 9

6. This matters if the software does not meet the manager's requirements at first.

earns: “if the software doesn't initially meet the manager's requirements (1)” Mark scheme 9626/13, November 2023, p. 9

7. The company will also own the copyright of the payroll software.

earns: “The company will own the copyright of the custom written software (1st)” Mark scheme 9626/13, November 2023, p. 9

8. This means it could sell the software to other businesses and earn money from it.

earns: “so can sell it to others (1)” Mark scheme 9626/13, November 2023, p. 9

A weak answer of the kind the examiners describe (written for this guide; likely about 2 of 8):

Custom software is made for the company, so it meets their needs. However, it is expensive and takes a long time to write, and it may have bugs because few people have tested it. Off-the-shelf software is cheaper and ready straight away.

Only the first sentence supports the decision. The drawbacks of custom software and the benefits of off-the-shelf software argue against the manager's choice, so they earn nothing.

“Most responses also gave arguments against the decision. These answers did not meet the demand of the question and were not worthy of credit.” Examiner report, November 2023, Paper 9626/13, p. 10

Every sentence argues for the manager's decision. Where a mark scheme shows (1st) then (1), state the benefit and then say what it leads to.

How much evidence: 20 Justify parts from 2022 onward (one, March 2026 Q10(c), has no matched mark scheme). The one-sided rule is repeated in many examiner reports from 2022 to 2025. The 'drawbacks of alternatives' exception rests on one examiner report sentence (June 2024, repeated for Papers 31 and 33), one 2023 examiner report comment and one mark scheme, so use it carefully.

Compare and Contrast

Compare — Cambridge's definition: "identify/comment on similarities and/or differences" Syllabus 2025–2027, p. 66

Contrast — Cambridge's definition: "identify/comment on differences" Syllabus 2025–2027, p. 66

Compare: similarities and/or differences. Give both unless the question says otherwise, because some questions require both. Contrast: differences only. Make every point clearly a comparison of the two things ('both', 'whereas', '-er than').

HOW THE MARKS ARE GIVEN

1. Mark schemes for Compare list similarities and differences, with one mark for each valid point. Some (for example March 2022 Paper 32, March 2023 Paper 32 and June 2022 Paper 12) say you need at least one similarity and one difference to get full marks.

"Must be at least one similarity and one difference for full marks" Mark scheme 9626/32, March 2022, p. 9

"Must be 1 of each for full marks" Mark scheme 9626/32, March 2023, p. 5

"Must have at least one similarity and one difference to gain full marks" Mark scheme 9626/12, June 2022, p. 4

2. Some mark schemes put a cap on each side, for example a maximum of 4 marks for similarities and 4 for differences in a 5-mark question. So writing only one side cannot get you full marks.

"Five marks available: Similarities Max 4" Mark scheme 9626/11, November 2023, p. 5

"Five from: Max 4 from: Similarities: Both" Mark scheme 9626/32, November 2025, p. 6

HOW TO WRITE IT

1. Write each difference as one matched sentence about both things, joined by 'whereas'. Do not write one paragraph about A and then a separate paragraph about B. Examiners did not award marks when they could not see the comparison.

"differences are best presented together, perhaps joined by the use of 'whereas'. In some cases, candidates wrote two separate descriptions one of phishing and the other of vishing." Examiner report, November 2022, Paper 9626/12, p. 3

"Where it was clear that these were presented as comparisons, marks were awarded if possible. However, this clarity was sometimes lacking and so marks were not awarded." Examiner report, November 2022, Paper 9626/12, p. 3

2. Do not answer with a bullet list or a table. One examiner report says lists of points, bulleted or in tables, are unlikely to get full marks for a comparison. Write full sentences.

"It should also be noted by centres that candidates who provide lists of points, bulleted or in tables, are unlikely to gain full marks because the question asks for comparisons which are best written in free response." Examiner report, June 2023, Paper 9626/32, p. 21

"Weaker answers just stated facts, sometimes as unconnected bullets, or described uses of LANs/WANs." Examiner report, March 2022, Paper 9626/32, p. 6

WHAT GOES WRONG

1. Students often forget the similarities. The November 2025 (Paper 11) and June 2025 (Paper 32) reports both noted that few candidates gave them.

"When candidates are asked to compare, they are able to identify similarities as well as differences. Few, if any, candidates identified similarities between the two methods." Examiner report, November 2025, Paper 9626/11, p. 2

"Most candidates could describe the basic, obvious differences between activity-on-node and activity-on-arrow PERT diagrams, but the similarities between the two were often omitted." Examiner report, June 2025, Paper 9626/32, p. 22

2. A true fact about only one thing, with no comparison shown, is not a comparison. 'A LAN covers a small area' gets no mark. 'A LAN covers a smaller area than a WAN' does. The June 2025 report (on a Contrast question) added that you need not spell out the other side every time, as long as the point is clearly a valid comparison.

"For example, 'a LAN covers a small geographical area' on its own is a true fact but does not answer the question. However, 'a LAN covers a smaller geographical area than a WAN' answers the question and would gain credit." Examiner report, March 2022, Paper 9626/32, p. 6

"whilst candidates do have to show some awareness that they are comparing, there is no requirement for candidates to present both sides of the argument." Examiner report, June 2025, Paper 9626/12, p. 6

3. Contrast means differences only. Similarities in a Contrast answer get no marks, so they waste your time.

"The command word 'contrast' requires only that the differences are described (see p. 45 of the syllabus). Any references to similarities were ignored and not given credit because these did not answer the question." Examiner report, June 2023, Paper 9626/32, p. 21

"When responding to 'contrast' questions, candidates should give only differences." Examiner report, March 2023, Paper 9626/32, p. 7

"Command word: Contrast: identify/comment on differences." Mark scheme 9626/32, March 2022, p. 3

4. Do not repeat the question as a similarity. If the question says both devices exchange data, 'both exchange data' earns nothing. Very general similarities, such as 'both are plastic cards', were also called weak.

"Centres should also remind their candidates that answers that repeat the question, in this case stating that both exchange data in networks, will not be given credit." Examiner report, June 2023, Paper 9626/32, p. 21

"the mark was for the repeatedly/iteration comment since 'loop through a block of code' was given in the question." Examiner report, November 2024, Paper 9626/32, p. 15

"Weaker answers referred to both being plastic cards and, inaccurately, both being used for online shopping." Examiner report, June 2023, Paper 9626/31, p. 17

5. Compare exactly the two things named in the question. Do not write about only one of them. Do not answer about something else that just has a similar keyword.

"Some candidates either only focussed on one of the two methods, or did not discuss the impact of each method on the quality of a video file." Examiner report, November 2025, Paper 9626/12, p. 7

"It was noted that a significant number of responses referred to dynamic IP assignment using DHCP. Such references did not answer the question" Examiner report, March 2023, Paper 9626/32, p. 8

TIPS

1. Give both similarities and differences in a Compare answer, even if the question does not say so. The examiners have not always agreed on this: some reports said both were required, and others said full marks were possible with only one side when the question did not ask for both. Writing both is safe in every case.

"Often a question will specify that both similarities and differences as required, as in this question, but even if it does not, candidates are advised to give both to ensure access to the full mark range." Examiner report, March 2023, Paper 9626/32, p. 8

"this is interpreted as both being required. Therefore, centres are reminded to ensure that their candidates include both. Answers that do not include both are unable to access the full range of marks." Examiner report, June 2023, Paper 9626/31, p. 17

"Where both similarities and differences, or where only similarities or only differences, are specifically required, this will be made clear in the question." Examiner report, November 2024, Paper 9626/32, p. 15

"However, in this case, full marks could have been achieved with only similarities or only differences in the answers." Examiner report, November 2022, Paper 9626/32, p. 18

WORKED EXAMPLE

Q3 [6 marks] “Compare the use of static and dynamic routing when sending data packets through IP networks. You must include similarities and differences.” Question paper 9626/32, March 2023, p. 4

A full-mark answer, built from the mark scheme (not a real candidate's):

1. Both static and dynamic routing are used in routers to decide the next hop that a data packet is sent to.

earns: “Used in routers/routing devices to determine next node/router/device/hop to send data packet/datagram/frame” Mark scheme 9626/32, March 2023, p. 5

2. Both types of routing provide a default route for a packet if no other route can be found.

earns: “Both provide default route for packet if no route can be determined” Mark scheme 9626/32, March 2023, p. 5

3. Static routing tables are created by network administrators, whereas dynamic routing tables are created by algorithms.

earns: “Static routing tables created by network administrators whereas dynamic routing tables are created by algorithms” Mark scheme 9626/32, March 2023, p. 5

4. Static routes are fixed, whereas dynamic routing lets packets take different routes depending on network conditions.

earns: “Static routes are fixed whereas dynamic routing allows packets to take different routes depending on network conditions” Mark scheme 9626/32, March 2023, p. 5

5. Static routing tables are not updated when network conditions or devices change, whereas dynamic routing tables are updated and can adjust to the changes.

earns: “Static routing tables not updated/amended when network conditions/device change whereas dynamic routing are updated/can adjust to changing network conditions” Mark scheme 9626/32, March 2023, p. 5

6. Static route information does not time out, whereas dynamic routes have a time-to-live after which the table is updated.

earns: “Static routing table/info does not ‘time out’ whereas dynamic routes can be time-limited/have time-to-live data” Mark scheme 9626/32, March 2023, p. 5

A weak answer of the kind the examiners describe (written for this guide; likely 0 of 6):

Static IP addresses are set by the network administrator and do not change. Dynamic IP addresses are given out by a DHCP server and can change each time a device connects.

It is about IP addresses, not routing. It also has no similarity, and it describes each type separately instead of comparing them.

“It was noted that a significant number of responses referred to dynamic IP assignment using DHCP. Such references did not answer the question and it can only be assumed that the candidates spotted the words ‘dynamic’ and ‘IP’ and misunderstood the question.” Examiner report, March 2023, Paper 9626/32, p. 8

Two similarities that start with 'Both' and four differences joined by 'whereas' give six separate marks. Each sentence covers both things, so the examiner can see the comparison.

WORKED EXAMPLE

Q1 [6 marks] “Contrast the characteristics of local area networks (LANs) and wide area networks (WANs).” Question paper 9626/32, March 2022, p. 2

A full-mark answer, built from the mark scheme (not a real candidate’s):

1. A LAN covers a smaller geographical area than a WAN.

earns: “LAN covers a smaller geographical area/valid examples than WAN” Mark scheme 9626/32, March 2022, p. 3

2. A LAN has a higher data transfer speed than a WAN.

earns: “LAN has higher data transfer speed/greater bandwidth than WAN” Mark scheme 9626/32, March 2022, p. 3

3. A LAN is usually owned by one organisation, whereas a WAN is often owned by a number of organisations.

earns: “LANs are (usually) owned by one organisation whereas WANs are (usually/often) owned by a number/collective ownership” Mark scheme 9626/32, March 2022, p. 3

4. LANs have fewer data transmission errors than WANs.

earns: “LANs have fewer data transmission errors than WANs” Mark scheme 9626/32, March 2022, p. 3

5. LANs have fewer network congestion problems than WANs.

earns: “LANs have fewer network congestion problems than WANs” Mark scheme 9626/32, March 2022, p. 3

6. A LAN uses Layer 1 or 2 devices such as switches and hubs, whereas a WAN uses Layer 3 devices such as routers.

earns: “LANs use Layer 1 or 2 devices/switches/bridges/hubs/repeaters whereas WANs use Layer 3 devices/Routers/Multi-layer Switches” Mark scheme 9626/32, March 2022, p. 3

A weak answer of the kind the examiners describe (written for this guide; likely 0 of 6):

- LAN: small area, one building.
- WAN: large area, the internet.
- Both are networks that connect computers.

The bullets are separate facts about one network at a time, not differences. The last bullet is a similarity, which gets no credit for Contrast.

“Weaker answers just stated facts, sometimes as unconnected bullets, or described uses of LANs/WANs. Others gave similarities. These did not gain credit because the responses did not answer the question.” Examiner report, March 2022, Paper 9626/32, p. 6

Every sentence names both LAN and WAN and uses a comparing word ('smaller ... than', 'whereas'). There are no similarities, because Contrast does not credit them.

How much evidence: Compare is well covered: the question papers have 16 question parts as Compare, all from 2022 onward, and many have examiner comments. Contrast has fewer: 8 parts are labelled Contrast. One of them, November 2024 Paper 33 Q3(b), is a false match ('contrast' is an image term in a Describe question), so 7 are real Contrast questions, 6 of them from 2022 onward (the other is June 2021 Paper 32). Another 20 parts from 2018-2021 are labelled 'Compare and Contrast'; they are not used here. The examiner reports disagree on whether Compare needs both sides when the question does not ask for both: June 2023 said both are required, while November 2022 and November 2024 said one side could get full marks. The advice to give both follows the March 2023 report and works under either reading.

Identify, State, Give, Define, Name, Suggest

Identify — Cambridge's definition: "name/select/recognise" Syllabus 2025–2027, p. 66

State — Cambridge's definition: "express in clear terms" Syllabus 2025–2027, p. 66

Define — Cambridge's definition: "give precise meaning" Syllabus 2025–2027, p. 66

Suggest — Cambridge's definition: "apply knowledge and understanding to situations where there are a range of valid responses in order to make proposals/put forward considerations" Syllabus 2025–2027, p. 66

Give — not in the syllabus list of command words. Not in the syllabus command-word table (page 66).

Name — not in the syllabus list of command words. Not in the syllabus command-word table (page 66).

List — not in the syllabus list of command words. Not in the syllabus command-word table (page 66).

Give exactly what is asked, in as many items as asked: the correct technical term or a precise fact. Do not copy words from the question, and do not explain unless the question also says describe or explain.

HOW THE MARKS ARE GIVEN

1. Short-answer questions usually give one mark for each correct item ('One from', 'Four from'). When the question says 'identify and describe' (or asks for an item and a description), the mark scheme can use matched pairs: one mark for the item and one for a description that goes with it.

"ONE from: • a cryptographic hash of the previous block • a timestamp • (all the) transaction data" Mark scheme 9626/33, June 2025, p. 7

"Eight available. Max two for each matched pair: • Table name/title (1st)" Mark scheme 9626/12, March 2024, p. 4

"Max FOUR marks, TWO per paired answer • Length check (1st)" Mark scheme 9626/12, June 2025, p. 10

WHAT GOES WRONG

1. Do not copy something from the question as your answer. If the question already names an item or gives an example, repeating it earns nothing. A rewording of the question earns nothing either ('a command line interface uses commands').

"Candidates should not repeat the question in their answers. For example, Question 5(a) asks for one 'other item of data' but a significant number of candidates included the items given in the question stem in their answer, which did not gain them credit." Examiner report, June 2025, Paper 9626/33, p. 24

"repetition of the theft of data, which was included in the question as an example, was not accepted." Examiner report, November 2025, Paper 9626/12, p. 6

"For example, candidates stated that a command line interface used commands or a gesture-based interface used gestures. In cases where candidates answered in these terms, without any further expansion, no credit could be awarded." Examiner report, November 2023, Paper 9626/12, p. 6

"Some candidates stated that the age of information meant that information was outdated, which was not accepted, as it was effectively, a repeat of the question." Examiner report, November 2024, Paper 9626/12, p. 3

2. A definition must be precise. Examiners praised short, exact definitions ('a sequence of characters'). They called nearly-right answers a common mistake, such as saying encrypted data is unreadable (it is readable, but its meaning is hidden). Learn the exact meaning of each syllabus term.

"Many candidates could give the precise definition of a string as a 'sequence of characters'." Examiner report, November 2024, Paper 9626/33, p. 18

"One common mistake is considering that encrypted data is unreadable without a key, which is one advantage of encrypting the data." Examiner report, June 2023, Paper 9626/11, p. 2

"Many candidates could not precisely define a virtual currency and muddled their answers with vague details of different forms of digital currency" Examiner report, November 2025, Paper 9626/32, p. 17

"This question assessed candidates' ability to provide definitions of key terms from the syllabus." Examiner report, June 2025, Paper 9626/13, p. 12

3. Use the exact names from the syllabus. Made-up names such as 'updating maintenance' got no marks. Be as specific as the question asks: 'numeric' was not enough when the question asked which numeric data type.

"Poorer answers incorrectly named the maintenance as e.g., 'updating maintenance' or 'replacement' maintenance. The names of the types of maintenance are given in Topic 16.8 so candidates are reminded to use these." Examiner report, November 2023, Paper 9626/32, p. 18

"Typical issues included stating that the field was numeric, but not stating what type of numeric field was best, or simply confusing the data types." Examiner report, November 2024, Paper 9626/12, p. 4

4. Answer the actual question. If it asks you to give reasons, a description gets no marks. If it asks how something could go wrong, a definition gets no marks.

"Many candidates described tunnelling which did not answer the question. The question required candidates to give reasons why it would be used." Examiner report, November 2022, Paper 9626/32, p. 16

"Candidates who failed to achieve marks usually did so because they gave a definition of visual checking, rather than answer the question." Examiner report, March 2025, Paper 9626/12, p. 2

5. Short answers still have to fit the scenario. Answers about the wrong part of the context were not accepted.

"Where candidates did not achieve full marks, this was often because they listed issues to do with the train service itself." Examiner report, November 2025, Paper 9626/12, p. 5

"When faced with a question that has a clear context, learners should take this as an opportunity to apply their understanding in the particular context." Examiner report, June 2023, Paper 9626/12, p. 6

TIPS

1. For Identify, State and Name, a single word or short term can be enough. Examiners described one Identify question as looking for four one-word answers. Another wanted only the name or acronym of a protocol. But when the question asks you to describe, one-word answers are not enough.

"In effect, this question was a test of memory recall, as whilst the answer did not require precise wording, it was looking for four one-word answers" Examiner report, June 2024, Paper 9626/11, p. 2

"This question required only a name or acronym of a protocol that works at the internet layer." Examiner report, June 2025, Paper 9626/31, p. 19

"As candidates were asked to describe, rather than state, answers needed to be more than one-word answers." Examiner report, November 2025, Paper 9626/11, p. 3

2. Give exactly the number of answers asked for. The November 2023 report says that only the first answers, up to the number asked, are marked, so extra answers earn nothing.

"Where candidates are asked for a specific number of answers, only the first of these up to that number will be marked. Therefore, for this question, if a candidate gave more than four answers, only the first four were marked." Examiner report, November 2023, Paper 9626/13, p. 8

3. Suggest is rare (2 parts from 2022 onward). For the one with an examiner comment (improving a flowchart), suggestions had to be relevant and placed clearly. A 2021 report on a Suggest question criticised answers that said what something was instead of giving reasons that fitted the scenario.

“As the question asks for improvement to the flowchart, candidates had to ensure that suggested improvements were both relevant and able to be clearly placed on the flowchart.” Examiner report, November 2023, Paper 9626/13, p. 10

“While many responses provided some detail about what incremental prototyping involved, most did not give reasons why it might be used when developing a spreadsheet for managing finances.” Examiner report, June 2021, Paper 9626/32, p. 20

WORKED EXAMPLE

Q2 [4 marks] “Identify four factors which can affect the quality of information.” Question paper 9626/11, June 2024, p. 2

A full-mark answer, built from the mark scheme (not a real candidate’s):

1. 1 Accuracy

earns: “Accuracy (1)” Mark scheme 9626/11, June 2024, p. 3

2. 2 Relevance

earns: “Relevance (1)” Mark scheme 9626/11, June 2024, p. 3

3. 3 Age

earns: “Relevance (1) Age (1)” Mark scheme 9626/11, June 2024, p. 3

4. 4 Completeness

earns: “Completeness (1)” Mark scheme 9626/11, June 2024, p. 3

The mark scheme accepts Accuracy, Relevance, Age, Level of detail and Completeness, one mark each. Four correct words get all four marks; the examiner report says the question was looking for four one-word answers. Write only four, because extra answers are not marked.

WORKED EXAMPLE

Q5(a) [1 marks] “Blockchains contain the records and the cryptographic hashes used to secure the records. State one other item of data that makes up the blockchain.” Question paper 9626/33, June 2025, p. 6

A full-mark answer, built from the mark scheme (not a real candidate’s):

1. A timestamp.

earns: “ONE from: • a cryptographic hash of the previous block • a timestamp” Mark scheme 9626/33, June 2025, p. 7

A weak answer of the kind the examiners describe (written for this guide; likely 0 of 1):

The cryptographic hashes that secure the records.

The question already gives this item. Repeating the stem earns nothing.

“a significant number of candidates included the items given in the question stem in their answer, which did not gain them credit.” Examiner report, June 2025, Paper 9626/33, p. 24

Read the stem and cross out anything already given before you answer. 'Other' means something not in the question.

WORKED EXAMPLE

Q1(a) [1 marks] “JavaScript can manipulate and store strings in variables. (a) Define the term string as used in JavaScript.” Question paper 9626/33, November 2024, p. 2

A full-mark answer, built from the mark scheme (not a real candidate’s):

1. A string is a sequence of characters.

earns: “(An object representing) a sequence of (any) characters.” Mark scheme 9626/33, November 2024, p. 5

One short, precise sentence gets the mark. The examiner report says the precise definition 'sequence of characters' was what good answers gave.

How much evidence: Counted from the question papers (from 2022 onward): Identify 28, State 12, Give 11, Define 8, List 2, Suggest 2, Name 1. The two List parts also ask you to describe, and the one Name part (March 2026 Paper 12 Q10) actually says 'identify'. Suggest rests on only one question from 2022 onward with an examiner comment (November 2023 question paper (Paper 13) Q8) plus one 2021 report, so treat that advice as thin. Give, Name and List are not in the syllabus command-word table. No examiner report we found says whether adding an example to a definition is rewarded or penalised, so this section makes no claim about it.

Complete, Draw, Write (algorithms and flowcharts)

Write (an algorithm / pseudocode) — not in the syllabus table of command words. Not in the command-word table (page 66). This is the syllabus topic 4.1 wording; the ellipses are part of the syllabus keywords.

“Writing an algorithm using pseudocode to solve a given problem, including: – INPUT/READ – WRITE/PRINT – FOR...NEXT – FOR...NEXT...STEP – IF...ELSE...ENDIF – WHILE...ENDWHILE – REPEAT...UNTIL – CASE...ENDCASE” Syllabus 2025–2027, p. 21

Complete (an algorithm or flowchart) — not in the syllabus table of command words. Not in the command-word table (page 66). Topic 4.1 wording.

“Writing and editing a given algorithm that demonstrates a decision-making process, including: – conditional branching – looping – nested loops – procedures/subroutines” Syllabus 2025–2027, p. 21

Draw (a flowchart) — not in the syllabus table of command words. Not in the command-word table (page 66). This is the syllabus symbol list on page 61 that examiners expect you to use.

“Flowchart symbols Element Symbol Input/output Decision Terminator (Start/Stop) Process box Subroutine Connector A Flow line” Syllabus 2025–2027, p. 61

Use the syllabus pseudocode keywords or flowchart symbols, set your variables to a starting value before the loop, put the counter update inside the loop, close every structure (ENDIF, ENDWHILE, UNTIL), and label every decision Yes and No.

HOW THE MARKS ARE GIVEN

1. Algorithm questions are marked one mark per required feature, such as initialising a variable, the loop, the input, the counter update and the output. Most features are marked separately, so you can earn marks even if part of your answer is wrong (a few marks depend on others). Always attempt the question: examiners say easier marks are generally there.

“Eight from: Input number of numbers Initialise count Initialise total Loop that works involving REPEAT... UNTIL (e.g. count = n) Input number within loop Increment count within loop Update total within loop” Mark scheme 9626/12, March 2022, p. 7

“Initialise Count before Mark variable is input (1) • REPEAT in correct logical location (1) • Input Mark (1)” Mark scheme 9626/13, June 2025, p. 7

“While questions about flowcharts and pseudocode often test a range of skills, easier marks are generally accessible, making it disappointing to see some candidates choose not to attempt this question.” Examiner report, June 2024, Paper 9626/13, p. 11

2. The counter must go up by one inside the loop, and not only inside an IF within the loop. The loop must stop at the right value.

“Correct Increment of Count (1) • Increment of Count is within the loop (1) • UNTIL Count = 20 (1) • UNTIL in correct position (1)” Mark scheme 9626/13, June 2025, p. 7

“count ← count + 1 (if within while statement but not if within IF statement (unless in both))” Mark scheme 9626/12, November 2023, p. 8

HOW TO WRITE IT

1. Indent the lines inside IF, ELSE and loops. Some mark schemes give a mark just for correct indentation. Variable names with two or more words must have no spaces. A slightly wrong variable name is not penalised if it is clear which variable you mean.

"IF and ENDIF indented correctly (1) • Use of THEN (1)" Mark scheme 9626/13, November 2023, p. 8

"Where marks were awarded, these tended to be for structural awareness, such as correct use of indentation, or for inputting a number at the start of the process." Examiner report, June 2024, Paper 9626/11, p. 4

"It should be noted that where variable names are made up of two or more words, there should be no gap between these words." Examiner report, November 2022, Paper 9626/12, p. 5

"(Ignore errors with variable names, as long as it is clear which variable is being referred to)" Mark scheme 9626/12, November 2022, p. 8

WHAT GOES WRONG

1. Initialise counters and totals before the loop starts, not inside it. In June 2025 Paper 13 Q7 this was the most common mistake among otherwise strong candidates, and in Q13 a missing initialisation was the usual reason for losing marks.

"most candidates realised that the algorithm required a count and that this count should be initialised outside of the loop. However, a sizeable minority placed the initialisation of the count within the loop." Examiner report, June 2025, Paper 9626/13, p. 11

"Where candidates did not achieve full marks, this was generally because they did not initialise the variable Sum." Examiner report, June 2025, Paper 9626/13, p. 12

"Two variables (e.g. MAX, COUNT, TOTAL, NUM) initialised/set to zero before loop (1)" Mark scheme 9626/12, November 2024, p. 5

2. Small syntax errors cost marks. Write each comparison in full (IF $X > Y$ AND $X < Z$, not IF $X > Y$ and $< Z$). Close every IF with ENDIF. Input or output each variable separately ('OUTPUT smallest and largest' was not accepted). Put text in speech marks, with commas between text and variables.

"treating the test criteria as a string and using the structure IF $X > Y$ and $< Z$, instead of IF $X > Y$ AND $X < Z$." Examiner report, June 2025, Paper 9626/13, p. 11

"Failure to close IF statements with ENDIF statements. • Failure to use mathematical reasoning in the pseudocode." Examiner report, June 2023, Paper 9626/12, p. 8

"For example, some responses stated, 'Output smallest and largest', which would not work and so was not accepted." Examiner report, November 2023, Paper 9626/12, p. 7

"Using an input statement 'input X and Y'. This would not correctly input the values for the two variables. • Not indenting within the IF statement and not closing the IF statement. • Not using speech marks and commas within the outputs" Examiner report, November 2023, Paper 9626/13, p. 10

3. In a flowchart, every decision box needs two exits labelled Yes and No (True and False are accepted). Write the condition as a maths expression, not words: 'is x between 5 and 10' is not acceptable.

"Candidates are reminded that all decision boxes must have a Yes and a No output (True and False are acceptable) and that they should use mathematical expression rather than text in their boxes. For example, the term 'is x between 5 and 10' is not acceptable." Examiner report, June 2023, Paper 9626/11, p. 4

"Firstly, decisions need to be based on questions, and each decision has two outcomes. These outcomes may be yes/no, or true/false, but they must be shown and labelled." Examiner report, March 2025, Paper 9626/12, p. 4

4. Use the symbols shown in the syllabus: parallelograms for input and output, rectangles for processes, and terminators for Start and Stop. Do not mix up input and process boxes, and do not use other versions of the symbols. Mark schemes give marks for correct shapes and for arrowheads on flow lines.

"Candidates need to use the symbols provided by the board and should have practiced using them correctly." Examiner report, March 2025, Paper 9626/12, p. 4

"In a small number of cases, candidates confused input symbols with process symbols" Examiner report, November 2024, Paper 9626/12, p. 4

"While it is accepted that there are variations on the use of symbols and their naming in system flowcharts, it is expected that for the purposes of this syllabus that the symbols shown in the syllabus be used." Examiner report, November 2024, Paper 9626/33, p. 20

"All lines have arrow heads with direction of flow 1 mark 9 All correct symbols/shapes 1 mark" Mark scheme 9626/13, June 2024, p. 7

TIPS

1. Use the structures and keywords from the syllabus (IF...ELSE...ENDIF, WHILE...ENDWHILE, REPEAT...UNTIL, CASE...ENDCASE). If the question names a structure, use it. Students who mixed CASE with IF statements, or who wrote prose, lost marks. Cambridge publishes a pseudocode guide that shows the expected layout.

"Performance on the algorithm question was improved, although some candidates did not use the coding structures and terms given for this qualification." Examiner report, November 2023, Paper 9626/11, p. 1

"Most candidates tried to combine the structure with IF statements, thereby highlighting this lack of understanding." Examiner report, June 2024, Paper 9626/11, p. 3

"proved challenging to candidates who wrote their answer in prose rather than using algorithmic constructs." Examiner report, June 2022, Paper 9626/11, p. 2

"Centres are reminded that CIE has produced a pseudocode guide for teachers and candidates working on this course." Examiner report, June 2025, Paper 9626/13, p. 11

2. Do exactly what the question asks. If it says 'redraw', draw a new flowchart rather than editing the printed one. If it asks for a flowchart, do not write pseudocode. Include every feature the question lists, such as a calculation and an output.

"In this question, candidates were clearly asked to redraw the flowchart. Candidates who amended the original flowchart and did not provide a new flowchart were therefore not answering the question." Examiner report, June 2025, Paper 9626/13, p. 12

"In a very small number of cases, candidates gave a form of pseudocode, instead of a flowchart for the answer" Examiner report, November 2024, Paper 9626/12, p. 4

"As a general comment on questions such as these, most candidates did not include all features identified in the question." Examiner report, November 2023, Paper 9626/11, p. 3

3. When you suggest or describe a change to a flowchart or pseudocode, say exactly where it goes, for example which line is removed and what replaces it. Write the answer in the answer space, not on the printed code.

"candidates need to be extremely clear about which line is being removed, and how it is to be replaced." Examiner report, June 2025, Paper 9626/11, p. 3

"centres are advised that candidates should answer the question in the space provided in the question paper, rather than writing answers on the pseudocode included in the question paper." Examiner report, November 2025, Paper 9626/12, p. 6

WORKED EXAMPLE

Q7 [8 marks] "Write a pseudocode algorithm using REPEAT...UNTIL which allows the teacher to input 20 test marks and outputs the correct message for each inputted mark." Question paper 9626/13, June 2025, p. 6

Marks above 70 get a Distinction, 55 to 70 a Merit, 40 to 54 a Pass, and below 40 'You have not reached the Pass mark'.

A full-mark answer, built from the mark scheme (not a real candidate's):

1. Count $\leftarrow$ 0 (before the loop)

earns: "Initialise Count before Mark variable is input (1)" Mark scheme 9626/13, June 2025, p. 7

2. REPEAT

earns: "REPEAT in correct logical location (1)" Mark scheme 9626/13, June 2025, p. 7

3. INPUT Mark

earns: "Input Mark (1)" Mark scheme 9626/13, June 2025, p. 7

4. IF Mark > 70 THEN PRINT "You have achieved a Distinction" ENDIF and IF Mark < 40 THEN PRINT "You have not reached the Pass mark" ENDIF

earns: "Correct Distinction grade criterion/correct fail criterion (1)" Mark scheme 9626/13, June 2025, p. 7

5. IF Mark > 54 AND Mark < 71 THEN PRINT "You have achieved a Merit" ENDIF and IF Mark > 39 AND Mark < 55 THEN PRINT "You have achieved a Pass" ENDIF

earns: "Correct Merit grade criterion/correct Pass grade criterion (1)" Mark scheme 9626/13, June 2025, p. 7

6. Count $\leftarrow$ Count + 1 (inside the loop)

earns: "Correct Increment of Count (1) • Increment of Count is within the loop (1)" Mark scheme 9626/13, June 2025, p. 7

7. UNTIL Count = 20 (as the last line of the loop)

earns: "UNTIL Count = 20 (1) • UNTIL in correct position (1)" Mark scheme 9626/13, June 2025, p. 7

8. Every PRINT message matches its range of marks.

earns: "All OUTPUTs correctly match the criteria (NOT awardable if BP4 or BP5 not awarded)

(1)" Mark scheme 9626/13, June 2025, p. 7

A weak answer of the kind the examiners describe (written for this guide; likely about 5 of 8):

```
REPEAT Count  $\leftarrow$  0 INPUT Mark IF Mark >= 55 and <= 70 THEN OUTPUT "You have achieved a Merit" ... Count  $\leftarrow$  Count + 1 UNTIL Count = 20
```

Count is reset to 0 on every pass, so the loop never ends and the initialisation mark is lost. The Merit test is not written in full (Mark > 54 AND Mark < 71), and the IF has no ENDIF. It can still pick up the REPEAT, INPUT, increment and UNTIL marks.

"However, a sizeable minority placed the initialisation of the count within the loop. This was the most frequently made mistakes amongst candidates who otherwise had a strong understanding of what was required." Examiner report, June 2025, Paper 9626/13, p. 11

This is pseudocode, so the strong answer is lines of code rather than sentences. The lines cover more than eight mark-scheme points (the increment and UNTIL lines earn two each), and the mark is capped at 8. The syllabus lists only the comparison operators >, < and =, and WRITE/PRINT for output, so the ranges are written with > and < (marks are whole numbers) and the output uses PRINT; the mark scheme's own example uses OUTPUT. Most of the marks are for structure: initialise before the loop, increment inside it, and use the right UNTIL condition in the right place.

WORKED EXAMPLE

Q4 [7 marks] “Draw a program flowchart to represent this algorithm: • Input an integer value. • Repeat this until you have entered 20 values. • At the end, output the largest value and the average of the values.” Question paper 9626/12, November 2024, p. 5

A full-mark answer, built from the mark scheme (not a real candidate's):

1. Start with a Start terminator and finish with an End (or Stop) terminator.
earns: “Start and End (accept STOP) terminators (1)” Mark scheme 9626/12, November 2024, p. 5
2. In a process rectangle before the loop, set COUNT and TOTAL to 0.
earns: “Two variables (e.g. MAX, COUNT, TOTAL, NUM) initialised/set to zero before loop (1)” Mark scheme 9626/12, November 2024, p. 5
3. Use parallelograms for every input of NUM and for the output of MAX and the average.
earns: “Input box and output box are both parallelograms (1)” Mark scheme 9626/12, November 2024, p. 5
4. Input the first NUM and set $MAX \leftarrow NUM$. Then, inside the loop, use process rectangles for $TOTAL \leftarrow TOTAL + NUM$ and $COUNT \leftarrow COUNT + 1$.
earns: “Sum all values input (1) 11 Set new MAX when input value > MAX in loop 12 Increment COUNT (1)” Mark scheme 9626/12, November 2024, p. 5
5. Draw two decision diamonds, 'NUM > MAX?' and 'COUNT = 20?', each with labelled Yes and No exits.
earns: “TWO Correctly drawn decision boxes with YES/NO decisions (1)” Mark scheme 9626/12, November 2024, p. 5
6. From 'NUM > MAX?', Yes goes to $MAX \leftarrow NUM$ and No skips it. From 'COUNT = 20?', No goes to an input box for the next NUM and back into the loop, and Yes leaves the loop.
earns: “All YES/NO decisions (used) go to correct outcomes (i.e. all logic is correct) (1)” Mark scheme 9626/12, November 2024, p. 5
7. After the loop, calculate $AVERAGE \leftarrow TOTAL / 20$ in a process rectangle.
earns: “Correctly calculate average (must include process/calculation) (1)” Mark scheme 9626/12, November 2024, p. 5

A weak answer of the kind the examiners describe (written for this guide; likely about 2 of 7):

A flowchart that puts 'Input number' in a rectangle, has a decision box 'Have 20 numbers been entered' with no Yes or No labels, and never sets COUNT or TOTAL to zero.

It uses the wrong symbol for input, writes the decision in words without Yes/No labels, and leaves out the initialisation before the loop.

“In a small number of cases, candidates confused input symbols with process symbols, but overall, performance on this question was very good.” Examiner report, November 2024, Paper 9626/12, p. 4

The mark scheme lists 14 creditable features for 7 marks, so getting the shapes, labels and initialisation right gives you room to spare. MAX starts at the first value rather than 0 so the answer is still right if every number is negative; the mark scheme also accepts setting MAX to 0. The weak answer is a description of a flowchart in words, because a drawing cannot be shown here.

How much evidence: Evidence is strong for pseudocode and program flowcharts in Paper 1 (many questions from 2022 onward with examiner comments). The June 2022 report says several times that algorithms and flowcharts were only recently introduced to the syllabus, so the most recent reports (2024-2025) are the best guide. Draw also covers DFD, PERT and system-flowchart questions in Paper 3; only the symbol points here apply to those. 'Write' is sometimes tagged on non-algorithm parts (for example November 2025 question paper (Paper 11) and November 2025 question paper (Paper 13) Q8(b), which are about audio), and those were ignored.

Where this comes from

Every statement in this guide is followed by the words it rests on, copied from a Cambridge document, with the page. There are 414 quotations from 72 documents: the 9626 syllabus for 2025–2027, the Cambridge Learner Guide for 9626, and the question papers, mark schemes and examiner reports for Papers 1 and 3 from 2017 to March 2026. A script checked each quotation word for word against the document it cites and wrote in the page number it found; the guide will not build if any quotation fails that check.

The worked examples use real questions. The full-mark answers were written for this guide by joining mark scheme points into sentences, and each sentence shows the mark scheme point it earns. The weak answers were also written for this guide, to show a mistake an examiner report describes. Neither is a real candidate's work.

What is not here: Cambridge's Example Candidate Responses booklets for 9626 (real scripts with examiner comments). They are on the School Support Hub, which needs a teacher login, so this guide does not quote them.

Limits: the share of marks per command word comes from automatic reading of the question papers. A part with no command word of its own counts under the word of the part before it. Papers 2 and 4 (practical) are not included.